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INDONESIA ENERGY OUTLOOK

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INDONESIA ENERGY OUTLOOK 2025



FOREWORD BY MEMBER OF THE NATIONAL ENERGY COUNCIL

Let us praise to God the Almighty for His blessing that the National Energy Council has completed the book of Indonesia Energy Outlook 2025.

Indonesia Energy Outlook potrays the energy in Indonesia including the current energy condition and future energy projection.

The book of Indonesia Energy Outlook 2025 is jointly written by Members of the National Energy Council from Stakeholders and Editorial Board at the Secretariate General of the National Energy Council with the assistance from the experts. The formulation of Indonesia Energy Outlook also involves related units in the Ministry of Energy and Mineral Resources namely the Data and Information Technology Center, Directorate General of Electricity, Directorate General of Oil and Gas, Directorate General of New Renewable Energy and Energy Conservation, as well as Directorate General of Mineral and Coal especially on the collection of current data and planning data.

Energy Outlook 2025 carries out 10 years ahead projection of the energy supply and demand in 2 (two) scenarios namely BaU scenario which depicts the existing policies and Green scenario which uses assumptions toward 2045 Advanced Indonesia and Net Zero Emission (NZE) by 2060.

In addition to this 2025 publication, Indonesia Energy Outlook also presents the result of projection based on 7 (seven) regions namely Sumatera, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku and Papua.

Jakarta, December 2025
Members of National Energy Council
Industrial Stakeholder

Abadi Poernomo

FOREWORD BY SECRETARY GENERAL OF THE NATIONAL ENERGY COUNCIL

The book of Indonesia Energy Outlook 2025 is an annual publication by the Secretariate General of the National Energy Council since 2014.

The energy supply and demand projection is conducted based on LEAP (Low Emissions Analysis Platform) modelling using 2024 as the baseline year taken from the Data and Information Technology Center MEMR, and BPS as well as data from related units.

In the effort to improve the quality of energy supply and demand projection data, Secretariate General of the National Energy Council would enhance the relationship with related parties to publish the book of Indonesia Energy Outlook as a reliable and trusted reference.

Jakarta, December 2025
Secretary General of the National Energy Council

Dadan Kusdiana

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- Members of the National Energy Council from stakeholders;
- Directorate General of Oil and Gas, Directorate General of Electricity, Directorate General of Mineral and Coal, Directorate General of New Renewable Energy and Energy Conservation, Data and Information Technology Center MEMR and Geological Agency;
- Danish Energy Agency;
- Purnomo Yusgiantoro Center;
- Energy experts for their contribution in formulating Indonesia Energy Outlook 2025.

DISCLOSURE

Indonesia Energy Outlook 2025 is an analysis on the long-term national energy demand and supply projection (2025-2035) with particular assumptions which are developed for the purpose of formulating future energy projection scenario. The assumption and projection are based on energy technology development including fossil and renewable energy based on current data and condition. The data in Indonesia Energy Outlook is taken from official publication as well as temporary data or data that are continuously updated by the sources.

EXECUTIVE SUMMARY

The 2025 edition of Indonesia Energy Outlook develops energy modeling for 2025 - 2035 into two scenarios, namely: the BaU scenario which uses current policy assumptions and the Green scenario which uses new policy assumptions towards a 2045 Advanced Indonesia and NZE in 2060. Projections are made in 7 (seven) regions, namely: Sumatera, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku, and Papua, taking into account the geographical conditions of several adjacent provinces.

The energy supply and demand analysis are conducted based on calculation of LEAP (Low Emissions Analysis Platform) model as a simulation model of energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population and household or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

As input in modeling, the main data sources used in formulating the 2025 IEO are the Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2024 published by Pusdatin MEMR, RUPTL PLN, RUKN, Statistik Indonesia 2023 - BPS, and BPS Industrial Statistics data. Regional energy consumption is determined by summing data from each province following the division per region. Energy consumption data per province is obtained from units in MEMR and Energy Business Entities. Specifically, fuel and LPG consumption data per region is obtained from fuel and LPG sales data per province from the Directorate General of Oil and Gas, for electricity consumption, the main data is based on PLN sales data, with additional identification of non-PLN consumption data and electricity of the smelter industry from the Directorate General of Electricity. For coal consumption data per province, it is obtained from the Directorate General of Mineral and Coal. Specifically, gas consumption data is based on sales data from HEESI, while consumption per industrial subsector uses BPS Industrial Statistics data. Household consumption data per province uses data from the Indonesia Residential End Use Survey which is the result of a collaborative study between CLASP and the Directorate General of NREEC.

The energy assumptions used in the BaU scenario refer to current conditions and future projections based on historical data from the last few years, including the implementation of biofuels, electric vehicles, etc. Meanwhile, the construction of power plants refers to the RUPTL. In the Green scenario, the economic and population growth assumptions are the same as the BaU scenario. However, in terms of energy utilization, assumptions are based on decarbonization efforts towards NZE 2060, both in the household, transportation, industry, and power plant sectors. As a baseline assumption, economic growth in both scenarios is 7.4%, while population growth is around 0.9%. For biodiesel in both scenarios, it is towards B-40. For bioethanol, the growth is around 5% in BaU scenario and 15% in Green scenario in 2035. The nuclear power plant in Green scenario will start in 2032 with a capacity of 250 MW and the use of hydrogen will begin in 2035 in the transportation sector, especially for buses and trucks.

Over the next 10 years, the BaU scenario projects an average annual increase in final energy demand of 6.1%, while the Green scenario projects an annual increase of only around 4.9% through the implementation of energy-saving and energy-conserving technologies across all sectors. The most striking differences between the scenarios are seen in the transportation and industrial sectors. In the BaU scenario, energy consumption in transportation grows by an average of 6% per year, while in the Green scenario it grows 4.2% per year. In the transportation sector, this decline is influenced by the adoption of electric vehicles which can save up to 70–80% more energy than fossil-fueled vehicles.

Total electricity consumption is projected to increase from 417 TWh in 2024 to 757 TWh (BaU) and 822 TWh (Green) in 2035 or experience an average growth of around 5.6% per year (BaU) and 6.4% per year (Green).

In line with the increase in electricity consumption, total electricity production is projected to increase to around 824 TWh (BaU) and 896 TWh (Green) from the fossil energy, especially coal. The share of electricity production from coal-fired power plants is projected to decline from around 61% in 2024 to 51% in the BaU scenario and 49% in the Green scenario in 2035. In contrast, electricity production from renewable energy plants in the next ten years, especially in the Green scenario, will experience significant growth of 18% per year. The largest increase in RE electricity production in the Green scenario comes from solar power with an average growth of 30% per year. Thus, the production increases from 1.4 TWh in 2024 to 26.1 TWh in 2035, supported by the continued declining trend in solar power plant costs.

To meet electricity demand, power plant capacity is projected to increase from 101 GW in 2024 to 196 GW (BaU) and 223 GW (Green) in 2035. All types of power plants experience positive growth, except for diesel power plants which continue to decline to only 6 MW in 2035. Renewable energy power plant capacity shows the highest growth compared to other energy sources. Thus, renewable energy power plant capacity will increase from 14 GW in 2024 to 31 GW (BaU) and 62 GW (Green) in 2035.

The total primary energy supply in 2035 is projected to increase by 6.7% and 5.7% per year respectively, to 543 million TOE (BaU) and 508 million TOE (Green). The renewable energy supply in 2034 reaches 98 million TOE under the BaU scenario, or approximately 18% of the total primary energy supply and 153 million TOE under the Green scenario, or 29% of the total primary energy supply.

In terms of emissions, total energy sector emissions in 2035 are projected to increase to 1,395 million tons of CO₂eq (BaU) and 1,110 million tons of CO₂eq (Green). In 2030, the emission achievements in both scenarios are 1,029 million tons of CO₂eq in BaU and 950 million tons of CO₂eq in Green or still lower than the emission target in the energy sector ENDC of 1,311 million tons of CO₂eq. Throughout the projection period, power plants remain the largest contributor to emissions due to the dominant use of fossil energy, especially coal, at around 35% in both scenarios in 2035. Meanwhile, the industrial sector contributes 34% in both scenarios. The high CO₂ emissions from industry happens since many equipment in the production, cooling, and heating processes remain using fossil energy and have not been substituted by electricity.

Indonesia is a country with low electricity consumption per capita. In 2024, Indonesia's average electricity consumption is 1,412 kWh per capita, far below the ASEAN average of 3,896 kWh per capita. By 2035, electricity consumption per capita will reach approximately 2,651 kWh/capita in the Green scenario and only 2,441 kWh/capita in the BaU scenario.

In 2024, Indonesia's final energy demand per capita is 0.67 TOE/capita, and in 2035 it is projected to increase to 1.17 TOE/capita (BaU) and 1.03 TOE/capita (Green). This figure is lower than the world's final energy per capita in 2020 reaching 1.32 TOE/capita based on IEA data in 2021.

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CHAPTER 01

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1. BACKGROUND

The book of IEO which is published annually is a result of an analysis that gives an overview of the national energy condition especially on energy supply and demand projection for the next 10 years (2025-2035).

The 2025-2035 projection uses two scenarios namely BaU scenario which uses current policy assumptions and Green scenario which uses new policy assumptions toward 2045 Advanced Indonesia and Net Zero Emission (NZE) 2060.

The main data source in IEO 2024 is Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2023 publication by Data and Information Technology Center MEMR, RUPTL PLN, RUKN, Statistics Indonesia 2023-BPS and Industrial Statistics of BPS.

1.2. METHODOLOGY

1.2.1. Modeling Analysis Framework

The modeling analysis is divided into three stages namely final energy demand analysis, energy transformation, and primary energy supply. Final energy demand analysis is done by using the assumptions of GDP growth, population growth, policy, strategic planning and roadmap on current energy development. The primary energy supply analysis also takes into account the use of various energy sources and energy resources potential including the current policies and current energy technology development. Meanwhile, energy transformation analysis is done by considering Electricity Supply General Plan (RUPTL) and National Electricity General Plan (RUKN). The modeling analysis framework is shown in Figure 1.1.

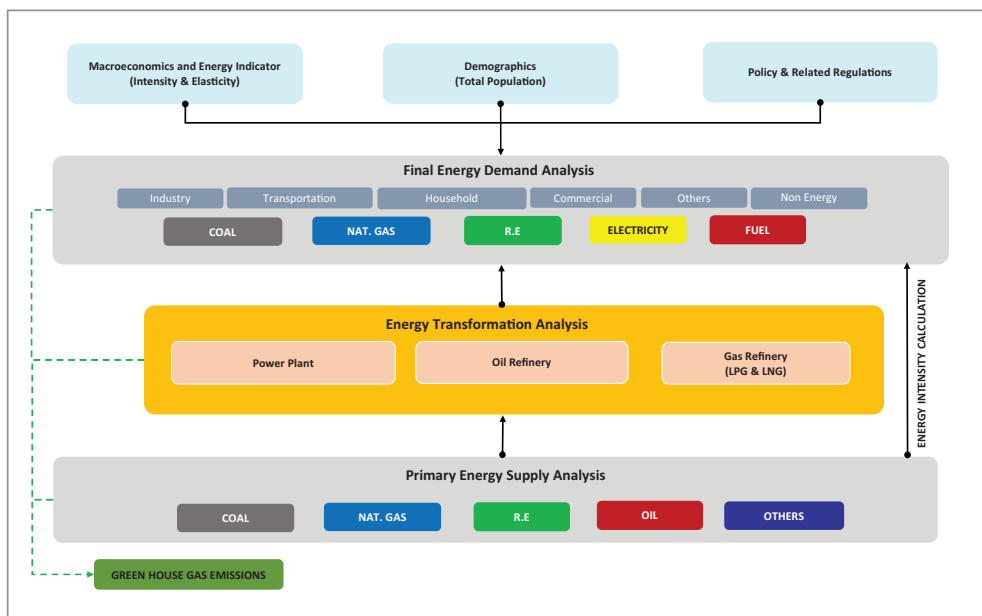


Figure 1. Modeling Analysis Framework

The energy supply and demand analysis are conducted based on calculation of LEAP model as a simulation model in energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population and household or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

Based on the modelling analysis framework in Figure 1.1, the considered parameters in final energy demand projection are social economic data (population and economic growth), historical data on energy consumption to identify energy intensity, and energy consumption pattern as the impact of people's lifestyle that is influenced by the projection of GDP increase or a more efficient technology.

In 2025, the calculation on energy supply demand projection per region is also carried out to obtain an overview on energy demand and supply in each region. The projection is done in 7 (seven) regions namely Sumatera region, Java-Bali region, Kalimantan

region, Sulawesi region, Nusa Tenggara region, Maluku region, and Papua region. The determination of 7 (seven) regions is considering the geographical condition of several regions that are located close to each other.

In the modeling, energy consumption uses the data on each province. It is then calculated based on per region. The data of energy consumption per province is obtained from units in MEMR and energy Business Entities. Data on fuel and LPG consumption/sales per province is specifically derived from Directorate General of Fuel and Gas. Meanwhile, the main data of electricity sales is derived from PLN sales data with additional identification of non-PLN consumption data and electricity in smelter industry from Directorate General of Electricity. Data on coal consumption per province is obtained from Directorate General of Mineral and Coal. However, the sales of electricity per industrial sub sector refers to data from Industrial Statistics BPS. Specifically for data on gas consumption is based on data on sales at HEESI, while consumption per industrial sub sector uses Industrial Statistics BPS data. The household gas consumption per province uses data from Indonesia Residential End Use Survey as the result of the joint study by CLASP and Directorate General of New Renewable Energy and Energy Conservation.

1.2.2. Energy Projection Scenario

The economic growth assumption used in both scenarios is around 7.4% and the population growth assumption is around 0.9%.

1.2.2.1. Business as Usual (BaU) Scenario

Nationally, the economic growth used in the projections is derived from BPS, LPEM UI and Bappenas 2024. These assumptions are also used to project energy demand in the current National Energy Policy based on Government Regulation No. 40 of 2025.

The projection of economic growth per province uses the GRDP Growth Projection data of 34 Provinces in 2021-2035 from Bappenas. To determine economic growth by region, summation and aggregation are carried out for each province.

Meanwhile, the population assumptions are from BPS publication (2020 Population Census). Similar to economic growth, population growth for each region also uses the aggregation method from each province.

The energy assumptions in BaU scenario refer to current conditions and future projections based on historical data from the last few years, including the implementation of biofuels, electric vehicles, and others, while the construction of power plants refers to RUPTL.

1.2.2.2. Green Scenario

In the BaU scenario, the assumptions of economic and population growth are similar to in the Green scenario. However, the energy utilization uses assumptions that lead to the 2045 Advanced Indonesia and 2060 NZE, including the following:

- a. Household sector: the use of electronic equipments that is projected to grow higher than in the BaU scenario as well as the conservation of equipment use in the household sector.
- b. Transportation sector: the use of EV and biofuel that is projected to be higher than in the BaU scenario (B40 by 2025 and E15 by 2035), the use of hydrogen by bus and truck in 2035, and energy conservation.
- c. Industrial sector: the gradual substitution from diesel as fuel in the generator to electricity, the substitution of coal to biomass, and unimplemented industrial conservation in the BaU scenario.
- d. Power plant: the addition of RE power plant capacity, especially Solar Power Plant, Wind Power Plant, and co-firing Coal Fired Power Plant that is higher than in the BaU scenario and the use of Nuclear Power Plant by 2032.

1.3. SYSTEMATIC

The systematic presentation of Indonesia Energy Outlook 2025 is as follows:

CHAPTER I INTRODUCTION

This chapter contains a brief explanation of the purpose and objectives of Indonesian Energy Outlook 2025, methodology, scenarios, and assumptions used in the modeling process.

CHAPTER II FINAL ENERGY CONSUMPTION OUTLOOK

This chapter presents the projections of Indonesia's energy conditions for the next 10 years along with the results of the analysis on final energy consumption per sector, final energy consumption per energy type, final energy per capita, electricity consumption per sector and electricity consumption per capita.

CHAPTER III POWER PLANT OUTLOOK

This chapter presents the projections of Indonesia's energy conditions for the next 10 years along with the results of the analysis on electricity production, renewable electricity production, power plant capacity and renewable energy power plant capacity.

CHAPTER IV OUTLOOK OF PRIMARY ENERGY SUPPLY AND CO₂ EMISSION

This chapter presents the projections of Indonesia's energy situation for the next 10 years, along with analysis on primary energy supply by type, primary energy per capita, and CO₂ emissions by sector, as well as an analysis of the emissions gap toward the Nationally Determined Energy Consumption (NDC). The chapter also discusses current performance in connection with the primary energy mix targets as stipulated in the Government Regulation on National Energy Policy.

CHAPTER V ENERGY PER REGION OUTLOOK IN BAU SCENARIO

This chapter presents the projection of Indonesia's energy conditions for the next 10 years along with the results of the analysis based on seven regions, namely Sumatera, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku, and Papua for the BaU scenario which includes final energy consumption per region, electricity consumption per region, electricity production per region, power plant capacity per region, renewable energy power plant capacity per region and primary energy supply per region.

CHAPTER VI ANALYSIS OF ENERGY TRANSITION ACHIEVEMENT

This chapter describes the progress of implementation of several national policies, which are primarily related to energy security and energy transition in Indonesia, which currently become the focus of National Energy Council, such as biofuels, hydrogen, nuclear power plants and electric vehicles.

CHAPTER VII CLOSING

This chapter contains conclusions from the recommendations of the analysis on energy supply and demand projections for the next 10 years.

CHAPTER 02

FINAL ENERGY CONSUMPTION OUTLOOK

CHAPTER 2

FINAL ENERGY CONSUMPTION OUTLOOK

2.1 FINAL ENERGY CONSUMPTION PER SECTOR

National energy demand is influenced by several key variables, such as economic and population growth, technological development, energy prices, infrastructure availability, and government policy direction. In the period of 2010 to 2024, Indonesia's final energy consumption is projected to increase by an average of 5.1% per year, from 94 million TOE in 2010 to 189 million TOE in 2024.

In 2024, the industrial sector becomes the largest final energy consumer with a share of 47%, followed by the transportation sector at 34%, households at 12%, commercial at 4%, and other sectors (agriculture, mining, and construction) at 1%. In addition to being the largest consumer, the industrial sector also experiences the highest growth in energy use with an average increase of 5.4% per year from 2010 to 2024. The high energy growth in the industrial sector is driven by the development of new industries, especially energy-consuming industries such as smelters. Currently, there are 54 nickel smelters in operation and 38 smelters under construction. Furthermore, the high growth rate is followed by the transportation sector (5.0%), the commercial sector (4.9% per year) and the household sector (4.7% per year). The increase in the transportation sector is driven by rapid urbanization, increasing community mobility, and the need for distribution of goods and economic activity.

Within the next ten years, provided there are no new policy interventions and market mechanisms remain the primary driver (BaU scenario), historical trends are expected to continue, resulting in high energy intensity. Under the BaU scenario, final energy consumption is projected to increase to 363 million TOE by 2035, or experiences an average annual growth rate of 6.1%.

Conversely, if the government strengthens the implementation of energy-efficient technologies through Best Available Technology (BAT) and encourages energy conservation across all sectors (the Green scenario), final energy consumption growth could be reduced to an average of 4.9% per year. By 2035, total final energy consumption under the Green scenario is estimated to reach 319 million TOE.

The most striking differences between the scenarios are seen in the transportation and industrial sectors. Energy consumption in transportation grows by an average of 6% per year in the BaU scenario and 4.2% per year in the Green scenario. In the transportation sector, this decline is driven by increased efficiency, primarily through the adoption of electric vehicles which can save up to 70–80% more energy than fossil-fueled vehicles. Meanwhile, in the industrial sector, the decline is driven more by conservation programs.

The development of final energy demand from 2010 to 2035 for the BaU and Green scenarios is shown in Figure 2.1.

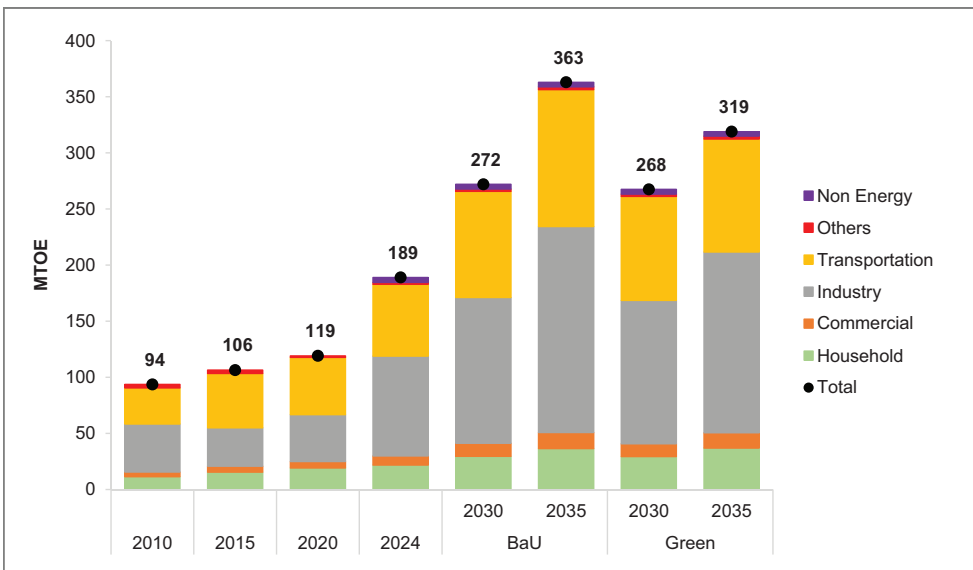


Figure 2.1 Final Energy Demand per Sector

Industry as the largest sector in final energy consumption is projected to increase by 6.8% per year in the BaU scenario (184 million TOE) and 5.5% per year in the Green scenario (161 million TOE) in 2035. In the Green scenario, there is a significant change in the share of energy use, especially for coal energy in which the share will decrease from 53.4% in 2024 to 48.5% in 2035 as an impact of the substitution of coal consumption

to biomass and electricity. Thus, the share of electricity and biomass will increase from 19.2% in 2024 to 19.8% in 2035 and 8% in 2024 to 10.2% in 2035, respectively. Efforts to substitute coal with biomass have been carried out in several industries such as the pulp and paper industry and the palm oil industry. Changes in the share of final energy consumption in the industrial sector are shown in Figure 2.2.

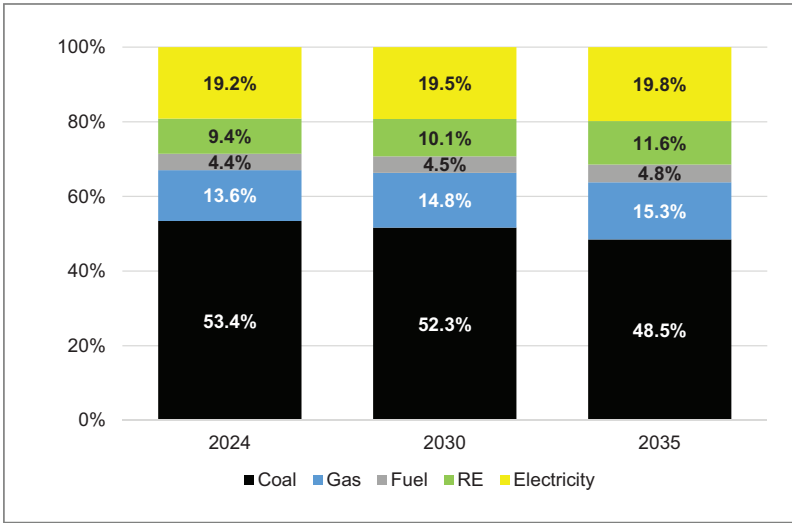


Figure 2.2 Final Energy Consumption Share in Industrial Sector

Approximately 30% of domestically available fuel is currently imported and the amount continues to increase due to the stagnation of fuel refinery capacity. Therefore, the government has implemented a policy of reducing the use of fuel. Diesel oil is currently being substituted with biodiesel and gasoline with bioethanol and electricity. With this policy, the share of fuel use will decrease from 83% in 2024 to 77% in 2035 in the Green scenario. Conversely, biofuel use will increase from 17% in 2024 to 23% in 2035.

Meanwhile, electricity consumption in the transportation sector with the share of only 0.1% in 2024 will increase to 0.8% in 2035. It is driven by the use of 5% or around 1.6 million units of electric cars, 5% or around 11.5 million units of electric motorcycles, and 2% or around 8,500 units of electric buses. In addition, in the Green scenario, it is projected that hydrogen vehicles will begin to be implemented with the share of 0.05% by buses and 0.01% by trucks. It means that their utilization contributes only 1,734 TOE. In addition, bioavtur is also assumed to begin being used for aviation with a mixture of 1% in 2035 reaching 0.43 MTOE. The overview of the final energy utilization in the transportation sector in the Green scenario is depicted in Figure 2.3.

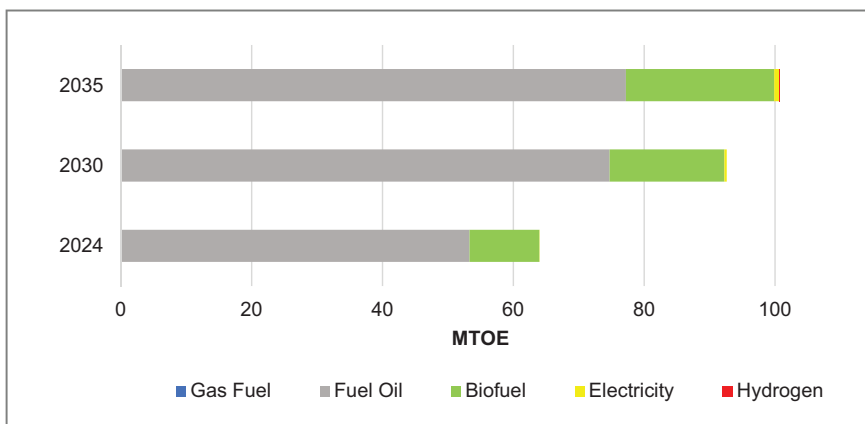


Figure 2.3 Final Energy Demand in Transportation Sector in Green Scenario

In 2024, energy consumption in the household was dominated primarily by electricity (53%) and LPG (46%), and the rest by gas, kerosene, and firewood. Over the next 10 years, electricity use will increase, primarily to replace LPG and due to lifestyle changes that increase the use of electronic appliances. Consequently, electricity demand will increase to 68% and LPG demand will decrease to 31% in the Green scenario by 2035 (Figure 2.4).

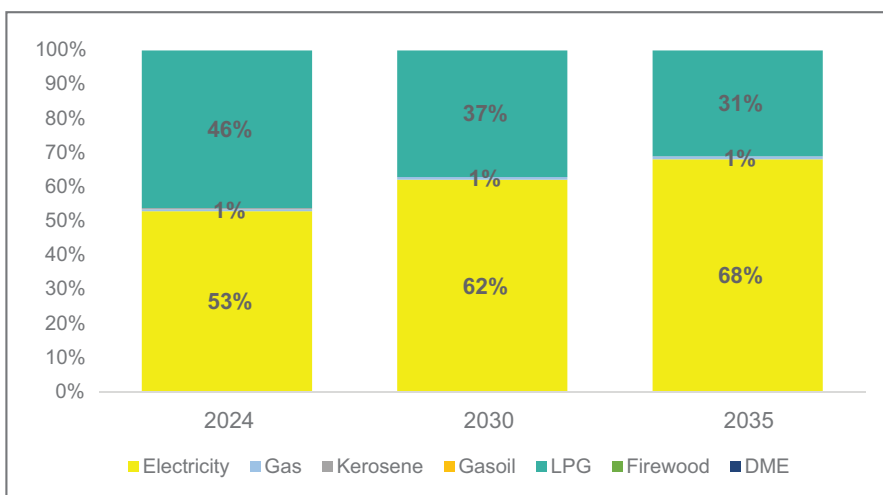


Figure 2.4 Share of Final Energy Demand in Household in the Green Scenario

In 2024, the commercial sector's energy consumption was dominated by electricity, accounting for approximately 88%, while the rest are by gas, LPG, biodiesel, and firewood. However, 10 years later, in the Green scenario, the share of electricity demand increases to 96%, while the share of other energy sources decreases, as shown in Figure 2.5.

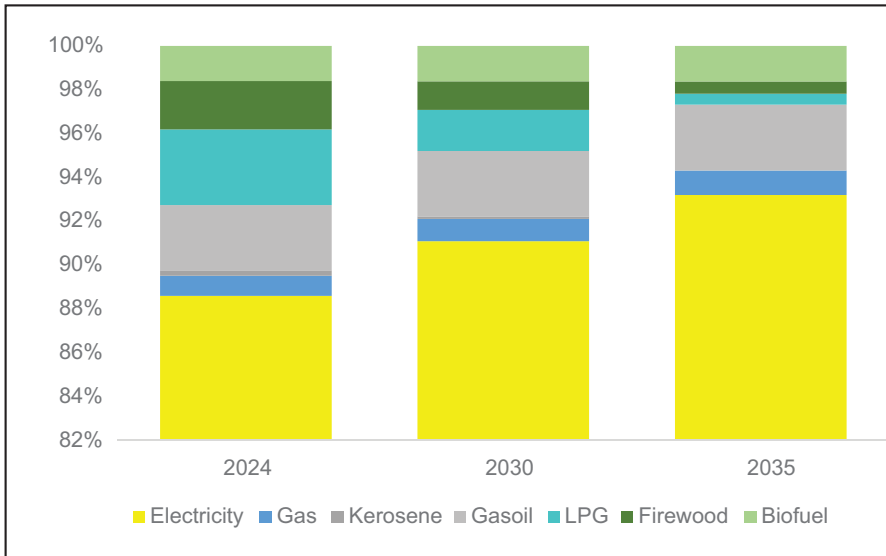


Figure 2.5 Final Energy Demand in Commercial Sector in the Green Scenario

2.2 FINAL ENERGY CONSUMPTION PER TYPE OF ENERGY

Until now, final energy consumption is still dominated by fossil energy with 71% share in 2024. Of the total consumption, the contribution comes from 31% (59 million TOE) of fuel, 25% (48 million TOE) of coal and 14% (27 million TOE) of gas. Meanwhile, electricity and RE contribute 19% (36 million TOE) and 10% (20 million TOE), respectively.

In the next ten years, despite that the contribution of fossil fuels is declining and their growth is below the average final energy consumption, fuel is estimated to remain the dominant energy source. It reaches, 31% in the BaU scenario with an average growth of 6.1% per year and 27% in the Green scenario with an average growth of 4.9% per year. The largest reduction in fuel consumption is contributed by the increasing substitution to electric vehicles as well as the expansion of biodiesel utilization and the utilization of hydrogen as a substitute for diesel oil.

For coal use, the BaU scenario still experiences the largest increase, namely 6.7% per year. Meanwhile, in the Green scenario, coal use growth is below average (4.6% per year). For gas, although its use is not as large as coal, it also experiences growth of around 3.8% in the Green scenario as gas is used to bridge the transformation from fossil energy to renewable energy. For renewable energy, both in scenarios, there is an increase of 6.6% per year in the BaU and 7.2% per year in the Green scenario. In the Green scenario, the highest renewable energy growth among other energy sources is influenced by the development of biofuels, especially the increase in 40% FAME (Fatty Acid Methyl Ester) mixture in diesel and the addition of 15% ethanol to gasoline in 2035 as seen in Figure 2.6.

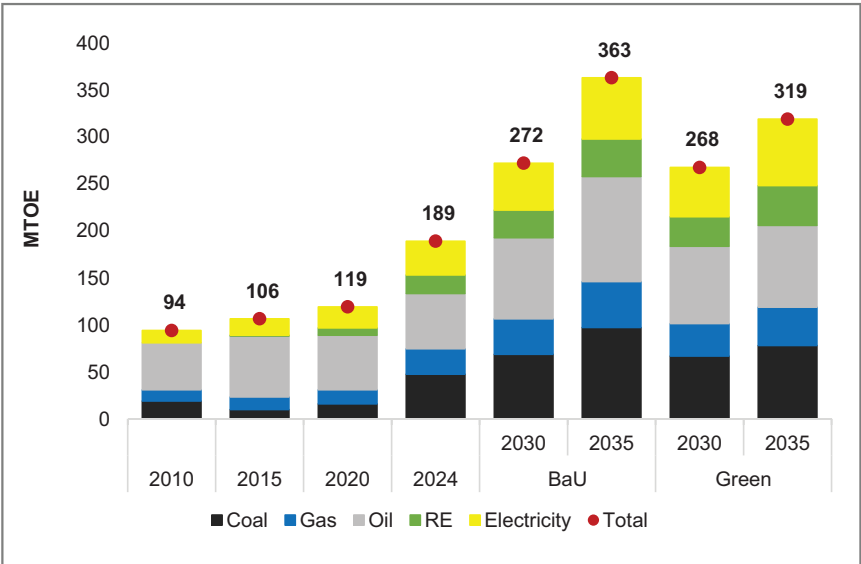


Figure 2.6 Final Energy Demand per Type of Energy

The share of final energy use specifically in the Green scenario shows that the role of fuel energy will decrease from 31% in 2024 to 27%, but the role of renewable energy will increase from 10% in 2024 to 13% in 2035. The development of the share of final energy consumption per type of energy is shown in Figure 2.7.



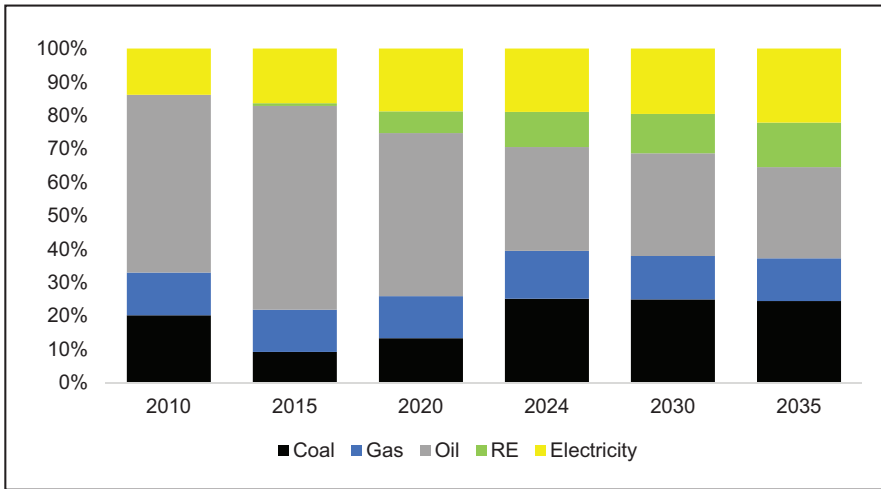


Figure 2.7 Final Energy Demand Share per Type of Energy in the Green Scenario

In 2024, Indonesia's final energy consumption per capita reached 0.67 TOE/capita and is projected to increase to 1.17 TOE/capita (BaU) and 1.03 TOE/capita (Green) in 2035. This value is still lower than the global average of 1.32 TOE/capita in 2020 (IEA, 2021). Compared to India, having the second largest population in the world, Indonesia's final energy consumption per capita is still lower, reaching 0.84 TOE in 2024. An overview of final energy consumption per capita for each scenario can be seen in Figure 2.8.

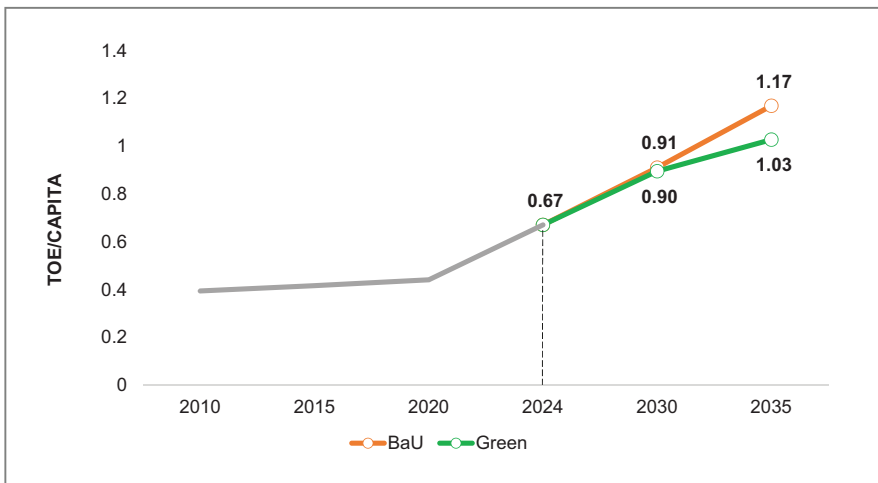


Figure 2.8 Final Energy per Capita National

2.3 ELECTRICITY CONSUMPTION PER SECTOR

National electricity consumption is projected to increase from 417 TWh in 2024 to 757 TWh in BaU scenario and 822 TWh in Green scenario in 2035. Thus, the average growth reaches 5.6% per year in the BaU scenario and 6.4% per year in the Green scenario.

In 2024, industrial sector is the largest electricity consumer with 48% share due to the large electricity consumption in industry, especially smelters which began to increase after 2021. In addition, electricity consumption in this study also includes captive electricity consumption outside the smelter that directly impacts to the quite significant increase in electricity demand. This condition will continue to dominate until the end of the projection year in 2035 with a share of around 46% in BaU scenario and 45% in Green scenario. Meanwhile, the household sector which dominated electricity consumption prior to 2020, will be in second place with a share of 32% in 2035. However, the consumption growth in the household sector is still higher than in the industrial sector, reaching 6.3% per year in the BaU scenario and 7.3% per year in the Green scenario. This occurs due to the continued increase in the use of electronic appliances in the household sector, such as AC, Refrigerator, Rice Cooker and other modern appliances in accordance with changes in lifestyle such as Air Fryer, Microwave, Food Processor, Blender, Chopper and others. Meanwhile, electricity consumption in industry grows by only 5.2% per year in BaU scenario and 5.9% per year in Green scenario.

The transportation sector is recorded as the sector with the highest growth in electricity consumption, namely 11% per year in BaU scenario and 33% per year in Green scenario although the total electricity consumption in 2024 is still low at 0.4 TWh. The figure is increasing to 1.3 TWh in BaU scenario and 9.5 TWh in Green scenario in 2035. This rapid growth is triggered by government programs to encourage the adoption of electric vehicles through various policies, including Government Regulation No. 55 of 2019 concerning the Acceleration of the Battery Electric Vehicle Program, Presidential Instruction No. 7 of 2022 concerning the Use of Battery Electric Vehicles as Operational Official Vehicles and/or Personal Vehicles of Central Government Agencies and Regional Governments, and electric motorbike purchase subsidy programs. The development of sectoral electricity consumption during the projection year can be seen in Figure 2.9.

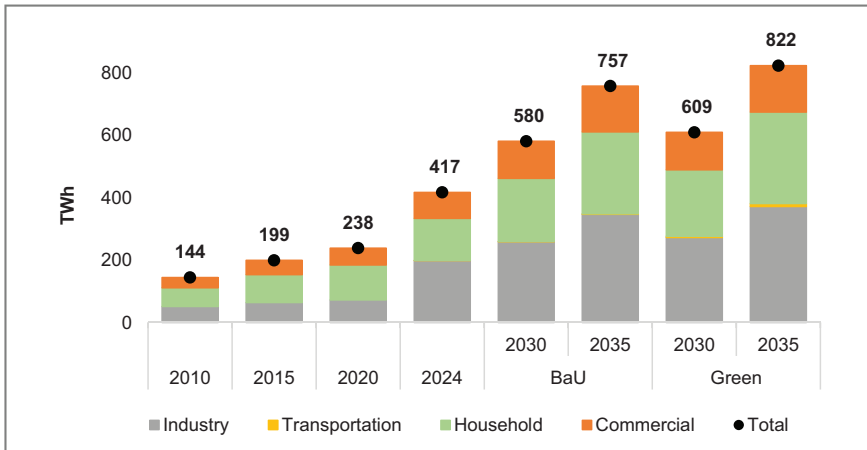


Figure 2.9 National Electricity Demand per Sector

2.4 ELECTRICITY CONSUMPTION PER CAPITA

Indonesia's electricity consumption per capita is still relatively low at 1,412 kWh/capita in 2024, far below the ASEAN average (3,672 kWh/capita).

Furthermore, in 2035, electricity consumption per capita is projected to reach 2,651 kWh/capita in Green scenario and only 2,441 kWh/capita per capita in BaU scenario. This low electricity consumption per capita is influenced by, among other things, the community's limited purchasing power and the low electricity consumption for industrial needs that use captive power, some of which data is not yet fully documented. The projection of electricity consumption per capita for each scenario can be seen in Figure 2.10.

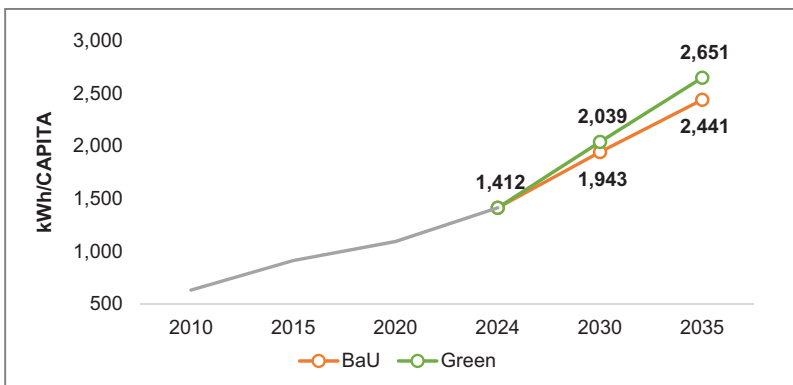


Figure 2.10 Electricity Consumption per Capita

CHAPTER 03

POWER PLANT OUTLOOK

CHAPTER 3

POWER PLANT OUTLOOK

Electricity demand continues to rise in line with economic growth, population growth, and people's lifestyles toward modernization. Electricity production is projected based on total electricity demand added by own use and losses during transmission, at around 15% in 2024 and 10% by 2035.

3.1 ELECTRICITY PRODUCTION

Electricity production in 2024 increases with an average growth of 5.8% per year since 2010 to reach 372 TWh. From 2010 until today, coal growth has increased the highest above average of around 9%, supported by the 35 GW program launched in 2015. The program currently has achieved around 27.8 GW. Similarly, the growth of electricity production from renewable energy power plants has continued to increase with a growth of almost 7% per year since 2010. Meanwhile, electricity production from gas power plants only grows 1.4%. On the contrary, electricity production from oil power plant is declining.

In the next 10 years, total electricity production is projected to reach 824 TWh (BaU) and 898 TWh (Green). The share of electricity from coal will decline from 61% in 2024 to 51% (BaU) and 49% (Green) in 2035. Conversely, electricity production from renewable energy power plants will increase from around 18% in 2024 to 28.5% in 2035 in the Green scenario, while the share in the BaU scenario remains the same. On the other hand, electricity production from diesel power plants is projected to be only 4.5 GWh in 2035 in the Green scenario. A complete overview of electricity production trends until 2035 is shown in Figure 3.1.

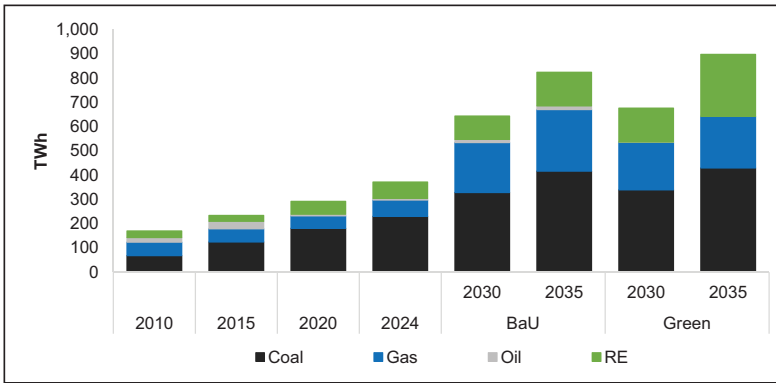


Figure 3.1 Electricity Production per Type of Energy

3.2 RE BASED ELECTRICITY PRODUCTION

Electricity production from renewable energy power plants includes electricity from hydro power plants (including mini-hydro and micro-hydro power plants), geothermal power plants, wind power plants, solar power plants, bioenergy power plants (biomass, biogas, and waste), and nuclear power plants. Electricity production from renewable energy power plants increases from approximately 67 TWh in 2024 to approximately 139 TWh in 2035 with 6.8% increase in the BaU scenario. However, in the Green scenario, electricity production from renewable energy power plants doubles compared to BaU, reaching 255 TWh. The largest growth in both scenarios comes from bioenergy power plants, in line with the increasing efforts of industrial sectors such as the pulp and paper and palm oil industries to use biomass as a substitute for coal. However, the largest share of renewable electricity production comes from hydro power plants and geothermal power plants, reaching more than 60% in both scenarios. The development and projection of renewable electricity production per type of power plant can be seen in Figure 3.2 below.

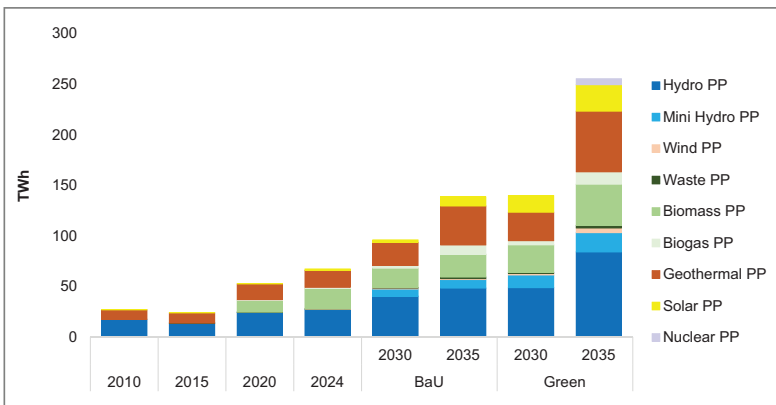


Figure 3.2 RE Electricity Production per Type of Power Plant

3.3 POWER PLANT CAPACITY

Indonesia's power plant capacity in 2024 reaches 101 GW. It grows 8% per year since 2010. The capacity growth is expected to reach around 11% per year in coal power plant, 6.1% per year in gas fired power plant, and only 1.5% per year in diesel power plant. Thus, the total share of fossil fuels power plant reaches 86% in 2024. Meanwhile, renewable energy power plant capacity growth is quite high, at 7.7% per year, but its share is only 14% of the total power plant capacity.

In the BaU scenario, in the next 10 years, there will be a significant increase in power plant capacity with a growth of 6.3% per year, reaching 196 GW with a renewable energy power capacity composition of around 16% in 2035 or around 31 GW.

In contrast, in the Green scenario, the power plant will grow by approximately 7.3% per year during the projection period, reaching 223 GW. Coal and gas power plant capacity will grow by approximately 6%, but diesel power plants will experience negative growth, leaving only approximately 6 MW in 2035. In line with emission reduction efforts, renewable energy power plants will experience significant growth of approximately 15.8% per year, reaching 62 GW in 2035. The development and projection of power plant capacity can be seen in Figure 3.3 below.

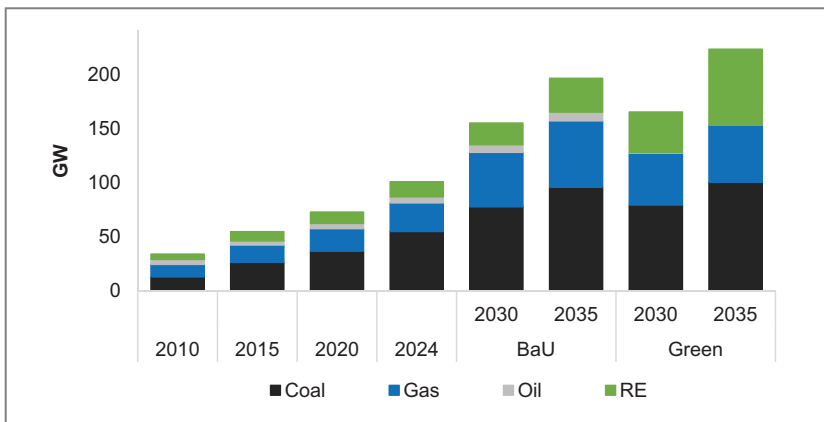


Figure 3.3 Power Plant Capacity per Type of Energy

3.4 RE POWER PLANT CAPACITY

Since 2010, hydro power plants, biomass power plants, and geothermal power plants have been the mainstays of RE power plant, with their capacities reached 7.1 GW, 2.9 GW, and 2.6 GW in 2024 respectively. Meanwhile, the total capacity of solar power plants, wind power plants, biogas power plants, and waste-to-energy power plants is very low at 1.3 GW.

In the Green scenario, RE power plant capacity will be almost twice as in the BaU scenario. Thus, by 2035, RE power plant capacity will reach 31.2 GW (BaU) and 61.6 GW (Green). A comparison of RE power plant capacity trends in the two scenarios can be seen in Figure 3.4 below.

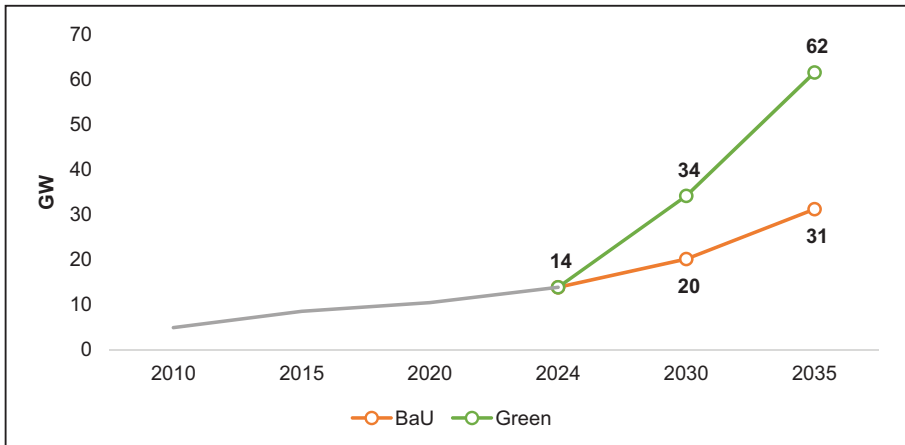


Figure 3.4 Trend of RE Power Plant Capacity per Scenario

Hydro power plant and solar power plants are the mainstays of RE power plant until 2035, accounting for more than 50% of the total RE power plant capacity. In line with this, hydro power plant capacity will increase from 7.1 GW in 2024 to 12.2 GW (BaU) and 22.8 GW (Green) in 2035. Meanwhile, solar power plants will increase from only 0.9 GW in 2024 to 6.5 GW (BaU) and 16.5 GW (Green). Such a massive increase in solar power plants is likely to happen considering that the price of solar panels is currently much cheaper than 10 years ago. Based on data from the International Renewable Energy Agency (IRENA), global crystalline solar PV module prices have fallen drastically between 2009 and 2023, with an average decrease of around 93%.

Furthermore, in the Green scenario, nuclear power plants are projected to begin the construction in 2032 with a capacity of 250 MW. The development and projected capacity of RE power plants can be seen in Figure 3.5 below.

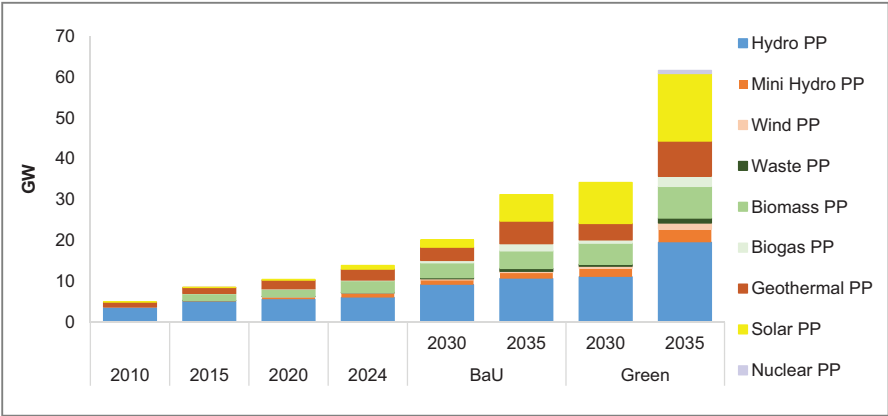


Figure 3.5 RE Power Plant Capacity per Type of Power Plant

CHAPTER 04

OUTLOOK OF PRIMARY ENERGY SUPPLY AND CO₂ EMISSION

CHAPTER 4

OUTLOOK OF PRIMARY ENERGY SUPPLY AND CO₂ EMISSION

4.1. PRIMARY ENERGY SUPPLY PER TYPE

Primary energy supply is divided into fossil and non-fossil energy. Fossil primary energy consists of coal, gas, and oil. Meanwhile, non-fossil primary energy includes various types of renewable energy (RE), namely hydro energy, solar energy, wind energy, geothermal energy, bioenergy, nuclear energy, hydrogen, and other renewable energies.

The trend in primary energy supply follows the same pattern as final energy consumption. From 2010 to 2024, primary energy supply will increase by 4.4% per year, from 151 million TOE (2010) to 276 million TOE (2024). Fossil fuel use remains dominant although its composition has changed. In 2010, oil accounted for 43% and coal for 26% to the composition. Meanwhile in 2024, the composition has changed into 40% of coal and 28% of oil. Gas supply has also decreased, from 25% (2010) to 17% (2024). In contrast, renewable energy supply has experienced significant growth at a rate of 12.1% per year, higher than the growth of coal supply, which is only 7.7% per year. Thus, the share of RE in total primary energy supply has increased sharply, from 5.4% in 2010 to 14.6% in 2024.

In the future, the total primary energy supply in 2035 will increase by around 6.7% per year (543 million TOE) in the BaU scenario and 5.7% per year (508 million TOE) in the Green scenario. In more detail, coal supply is projected to grow at the average of 5.8% per year in the BaU scenario and slow down of about 4.0% per year in Green scenario. The decline in coal use in the Green scenario is due to the co-firing program in several coal-fired power plants whose mixture continues to increase from the current 3.5% to 10% by 2035. Like past trends, the use of new and renewable energy will continue to be encouraged and even prioritized. Thus, the supply of primary energy from RE will grow

the highest by around 8.4% (BaU) and 12.9% (Green). This is particularly influenced by the increase in the capacity of RE power plant, especially Solar Power Plant and the program to increase the use of B-40 biofuels and B-15 bioethanol by 2035. The projection of primary energy supply per type for both scenarios can be seen in Figure 4.1.

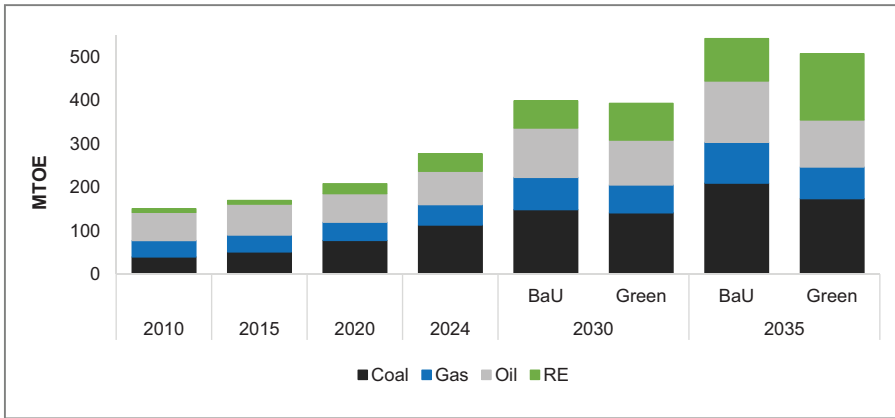


Figure 4.1 Primary Energy Supply Projection per Type of Energy

In 2035, renewable energy supply is estimated to reach 98 million TOE (18% of total primary energy supply) in the BaU scenario and 153 million TOE (29% of total primary energy supply) in the Green scenario. The primary energy mix in 2035 for both scenarios can be seen in Figure 4.2 and 4.3.

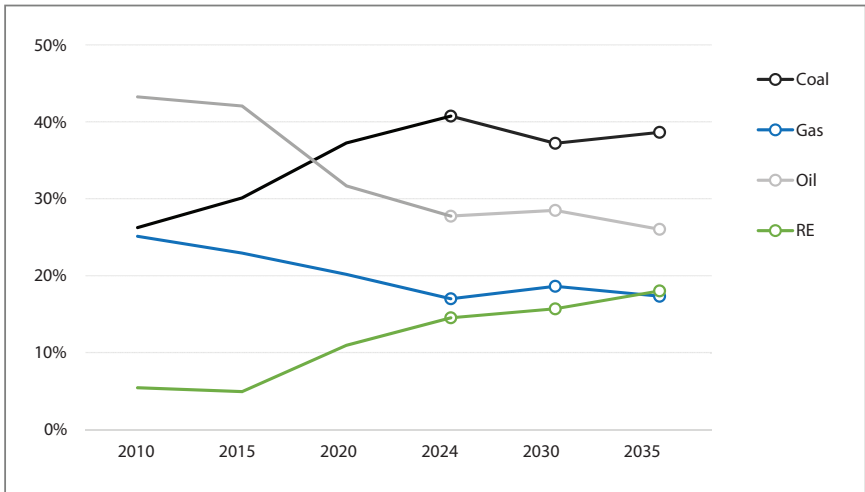


Figure 4.2 Primary Energy Mix in BaU Scenario

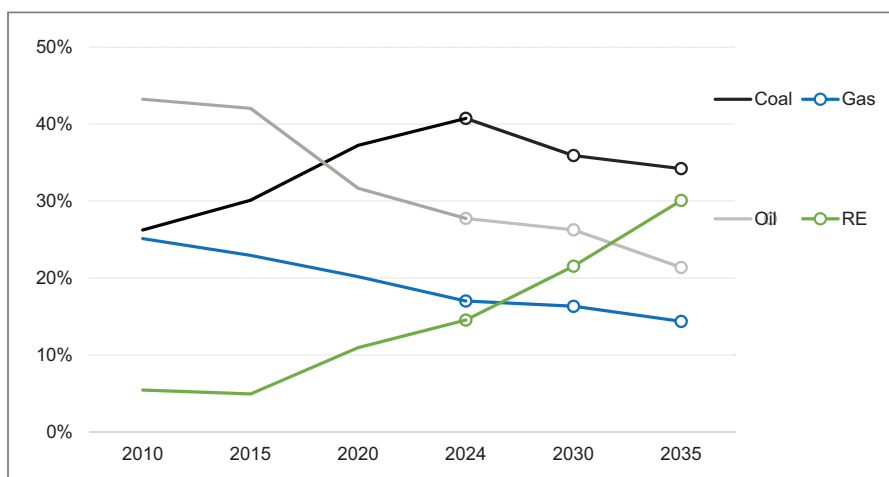


Figure 4.3 Primary Energy Mix in Green Scenario

4.2 PRIMARY ENERGY SUPPLY PER CAPITA

In 2035, primary energy supply per capita is estimated to reach 1.8 TOE per capita in the BaU scenario and 1.6 TOE per capita in the Green scenario. This figure is lower than the global average in 2020, which reached 1.70 TOE per capita (IEA, 2021). Projections of the primary energy supply per capita national for each scenario can be seen in Figure 4.4.

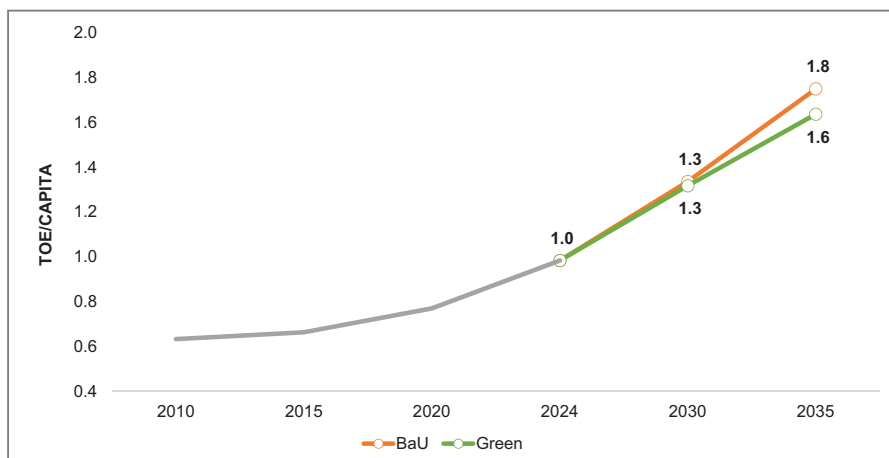


Figure 4.4 Primary Energy Supply per Capita

4.3 CO₂ EMISSIONS PER SECTOR

When discussing greenhouse gas (GHG) emissions in the energy sector, the use of fossil fuels can be separated as they are the primary contributor to GHG emissions, particularly CO₂. Therefore, it is certain that the greater use of fossil fuels without offsetting emissions capture or control technologies, the greater the resulting emissions will be in line with that energy use.

As explained previously, until now and in the next few years, fossil fuels will still be dominating the energy use in Indonesia. In the period of 2010 to 2023, CO₂ emissions increased from 421 million tons CO₂-eq (2010) to 839 million tons CO₂-eq (2024), with an average growth rate of 5.06% per year. The industrial and power plant sectors are the largest contributors to emissions, contributing 31% and 30% respectively in 2010 and 27% and 37% in 2024. In addition to these two sectors, the transportation sector also has a significant contribution, namely 23% in 2010 and 19% in 2024.

Total emissions in 2035 are estimated to increase to 1,395 million tonnes CO₂-eq (BaU) and 1,110 million tonnes CO₂-eq (Green). In 2030, emissions from both scenarios (1,029 million tonnes CO₂-eq in the BaU scenario and 950 million tonnes CO₂-eq in the Green scenario) are still lower than the energy sector emission target in the ENDC at 1,311 million tonnes CO₂-eq .

During the projection period, industrial sector emissions grew the highest compared to other sectors, namely 7% (BaU) and 5% (Green). Thus in 2035, the emissions will reach 474 million tons of CO₂-eq and 384 million tons of CO₂-eq respectively. In addition, the industrial sector is also the largest contributor to emissions due to the high dominance of fossil energy use in this sector, especially coal which contributes around 35% in both scenarios. The high CO₂ emissions from the industrial sector are caused by the large number of equipment that still uses fossil energy in the production, cooling, and heating processes, which cannot yet be replaced by electricity.

In power plants, the emission trend continues to increase by 5.4% (BaU) and 3.2% (Green), respectively. Thus, by 2035, emissions in the power plant sector will reach 572 million tons of CO₂-eq (BaU) and 454 million tons of CO₂-eq (Green). The low emission growth in the Green scenario is influenced by the higher number of RE power plants built by the end of the projection year.

The transportation and household sectors also contribute to emissions reductions in the Green scenario that the average emissions growth is reduced to 3.5% and 1.2%, respectively. Thus, total emissions in the Green scenario are 20% lower compared to in the BaU scenario. The development of GHG emissions in CO₂-eq from 2010 to 2035 by energy-consuming sector can be seen in Figure 4.5.

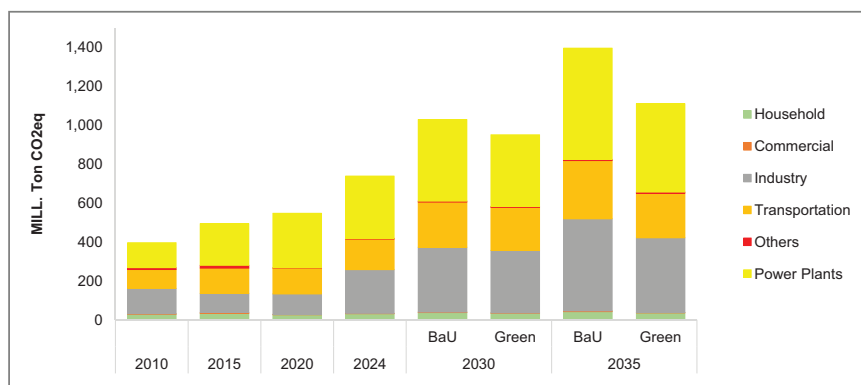


Figure 4.5 Greenhouse Gas Emission 2010 – 2035

4.4 ANALYSIS OF THE GAP IN EMISSION ACHIEVEMENT AGAINST NDC (ENDC)

Within the Enhanced Nationally Determined Contribution (ENDC) framework, Indonesia is committed to reducing greenhouse gas (GHG) emissions by 358 million tons of CO₂-eq by 2030 through independent efforts or 446 million tons of CO₂-eq with international support. The emission targets by 2030 are 1,311 million tons of CO₂-eq through independent efforts and 1,223 million tons of CO₂-eq with international assistance.

In this 2025 Outlook, in the BaU Scenario, GHG emissions are projected at 1,029 million tons of CO₂-eq in 2030. Meanwhile, in the Green Scenario, they are projected at 950 million tons of CO₂-eq.

Compared to the unconditional target (1,311 million tons of CO₂-eq), emissions in the BaU scenario provide a positive gap of 282 million tons of CO₂-eq. It means that Indonesia has been able to achieve the unconditional target set in the ENDC merely with current policies and without additional transformative policies. Meanwhile, the Green scenario produces a positive gap of 361 million tons of CO₂-eq, exceeding the unconditional ENDC target. This shows that the implementation of more aggressive policies in the use of renewable energy and energy efficiency provides a much greater emission reduction than the target.

Furthermore, when the emission projections are compared with the conditional target (1,223 million tons of CO₂-eq), the BaU scenario shows a gap of 194 million tons of CO₂-eq, which still indicates the achievement of the target. However, this achievement is minimal. It also indicates that without more decisive policy measures, Indonesia will almost reach it simply by maintaining the existing trend. Meanwhile, the Green scenario has a gap of 273 million tons of CO₂-eq, exceeding the more moderate conditional target. These results indicate that with the right policies, Indonesia will not only meet the conditional ENDC target but will also achieve the target with a significant surplus.

The BaU scenario offers an optimistic view that Indonesia can achieve its ENDC targets without major interventions, but this will only represent a minimum achievement. The gap of 282 million tons of CO₂-eq against the unconditional target and 194 million tons of CO₂-eq against the conditional target indicates that while Indonesia can meet its targets without major policies, this achievement leaves no room for the more aggressive reductions that are crucial to avoid worse climate change impacts in the future. In this scenario, Indonesia remains highly dependent on existing energy market trends, which are largely driven by fossil fuel consumption. Therefore, the risk of dependence on fossil fuels and the potential inability to achieve long-term targets remain key issues that must be addressed.

The Green scenario illustrates Indonesia's potential in creating significant structural changes in energy consumption, prioritizing energy-efficiency technologies, increasing renewable energy capacity, and transitioning to electric vehicles and biofuels. With a projection of 950 million tons of CO₂-eq, this scenario shows a larger gap of 361 million tons of CO₂-eq against the unconditional target and 273 million tons of CO₂-eq against the conditional target. These results indicate that if emissions reduction policies are implemented more intensively with the support of investments in clean technology and energy efficiency, Indonesia has the capacity to not only meet but exceed the ENDC targets. It also demonstrates that strategic sectors, such as transportation and industry, can play a significant role in reducing emissions.

Despite that the Green scenario demonstrates higher performance, a significant challenge occurs in implementing policies that support this energy transition. The adoption of electric vehicles and renewable energy requires strong policy support, adequate infrastructure, and incentives for industry and society. The transition to a green economy requires transformation of the industrial, energy, and transportation sectors, as well as policy reforms involving all stakeholders, from the government, the

private sector, to the public sector. Fiscal policies such as clean energy subsidies and investment incentives will be crucial in accelerating the achievement of low-carbon emissions. On the other hand, while the BaU scenario can meet emissions targets, the dependency on market policies without active government support will be risky in leaving Indonesia unprepared to face the long-term challenges of climate change and energy security.

CHAPTER 05

ENERGY OUTLOOK PER REGION OF BAU SCENARIO

CHAPTER 5

ENERGY OUTLOOK PER REGION OF BAU SCENARIO

5.1 REGIONAL FINAL ENERGY CONSUMPTION

The Java–Bali region still dominates national final energy consumption, with a share of approximately 54% in 2024 and is projected to decline to approximately 41% in 2035. This dominance is in line with Java–Bali’s GRDP contribution of 65% of total national GDP and its population proportion of approximately 57% in 2024 (BPS, 2024). The decline in the Java–Bali region’s share causes the average growth rate of energy consumption in this region to be relatively lower compared to other regions, as well as nationally.

Based on the projections, the final energy consumption share of the Maluku region is only around 2.1% in 2024 and 3.2% in 2035, but it shows the highest consumption growth reaching around 10.5% per year. This increase is mainly influenced by the development of the mineral processing and refining industry in North Maluku Province as a consequence of government policies that encourage the downstreaming of natural resources. Based on data from the North Maluku Industry and Trade Office, there are currently seven smelter industries spread across North Maluku Province as shown in Table 5.1.

Table 5.1 Smelter Industries in Maluku Region

No	Name of Company	Industrial Zone
1	PT Indonesia Weda Bay Industrial Park	Weda Bay Industrial Zone
2	PT Kao Rahai Smelters	Weda Bay Industrial Zone
3	PT Jiu Long Metal Industry	Weda Bay Industrial Zone
4	PT Halmahera Jaya Feronikel	Obi Industrial Zone
5	PT Halmahera Persada Lygend	Obi Industrial Zone
6	PT Megah Surya Pertiwi	
7	PT Weda Bay Nickel	Obi Industrial Zone

Source: Government of North Maluku Province

A similar phenomenon happens in Sulawesi region, with final energy consumption growth estimated at 10.3% per year. Similar to Maluku region, the growth in final energy consumption in Sulawesi region is driven by the construction of smelter facilities. Based on data from the Regional Development Planning Agency and the Energy and Mineral Resources Offices in the provinces of South Sulawesi, Southeast Sulawesi, and Central Sulawesi, there are currently 19 mineral processing industries in the Sulawesi region, as shown in Table 5.2 below.

Table 5.2 Smelter Industries in Sulawesi Region

No	Company Name	Province
1	PT. Sulawesi Mining Investment	Central Sulawesi
2	PT. Ghuang Cing	
3	PT. Indonesia Tsingshan Stainless Steel	
4	PT. Tsingshan Steel Indonesia	
5	PT. Indonesia Morowali Industrial Park	
6	PT. Walsin Nickel Industrial Indonesia	
7	PT. Gunbuster Nickel Industri	
8	PT. Lestari Smelter Indonesia	
9	PT. Hua Chin Aluminum Indonesia	
10	PT. Oracle Nickel Indonesia	
11	PT. Qing Feng Ferrochrome	
12	PT. WXNI	
13	PT. Indonesia Morowali Industrial Park	
14	PT. Ceria Nugraha Indotama di Kab. Kolaka	South East Sulawesi
15	PT. Vale Indonesia di Kab. Kolaka	
16	PT. Artha Mining Industry di Kab. Bombana	
17	PT. Virtue Dragon Nickel Industry di Kab. Konawe	
18	PT. Macika Mineral Industri di Kab. Konawe Selatan	South Sulawesi
19	Vale Indonesia	

Source: Government of South Sulawesi, Southeast Sulawesi, and Central Sulawesi

A visual overview of final energy consumption per region can be seen in Figure 5.1.

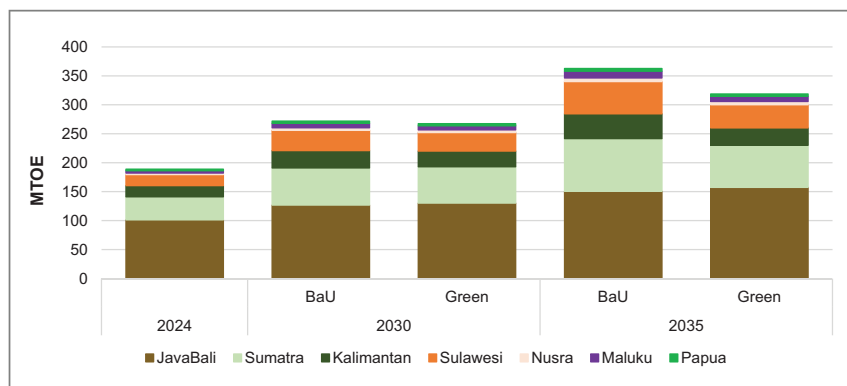


Figure 5.1 Final Energy Consumption per Region

5.2 REGIONAL ELECTRICITY CONSUMPTION

By 2024, the Java-Bali region was recorded as the largest electricity consumers, of around 45% of the total national consumption, followed by Sumatera (21%), Sulawesi (16%), Maluku (7%), Kalimantan (6%), Nusa Tenggara (2%), and Papua (3%). This distribution reflects the pattern of population distribution and economic activity which is still concentrated in Java-Bali. Although the area of Papua is much larger than Java (around 6 times larger), its electricity consumption is 13 times lower than Java-Bali, because the population in the Papua region is only 4% of the population in Java and Bali. In addition, Papua's GDP is only 3.4% of Java and Bali's GDP.

In the next 10 years, in the BaU scenario of electricity consumption in the region, Java-Bali remains the main contributor to electricity demand, but its share will decrease from 45% in 2024 to 40% in 2035. It is in line with the increasingly widespread distribution of electricity network infrastructure outside Java, including the 1 MW program in village and remote areas that will also cover regions outside Java-Bali where the electrification ratio is less than 100%. An overview of the trend of projection on electricity consumption per region can be seen in Figure 5.2.

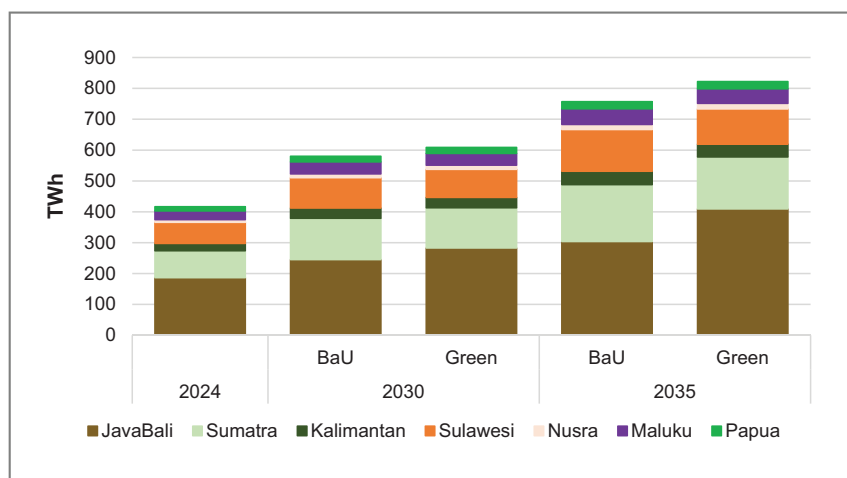


Figure 5.2 Regional Electricity Consumption

5.3 REGIONAL POWER PLANT CAPACITY

Power plant development will be adjusted to the increase in electricity consumption in each region. Until 2035, the largest power plant capacity will remain in the Java-Bali region, but its share will decline from 54% in 2024 to 46% in 2035. Conversely, the share of generating capacity in non-Java-Bali regions will increase, except in Sumatra and Papua.

In the Kalimantan region, the growth in power plant capacity in 10 years will reach 12.5% per year which is driven by, among others, the increase in the capacity of the Sei Mentarang Hydro Power Plant in North Kalimantan by 1,350 MW which will be used for the bauxite processing industry and the construction of a 50 MW Solar Farm in the IKN as well as the development plan of the Ministry of Communication and Information's Data Center for the National Data Center.

A complete overview of regional power plant capacity can be seen in Figure 5.3.

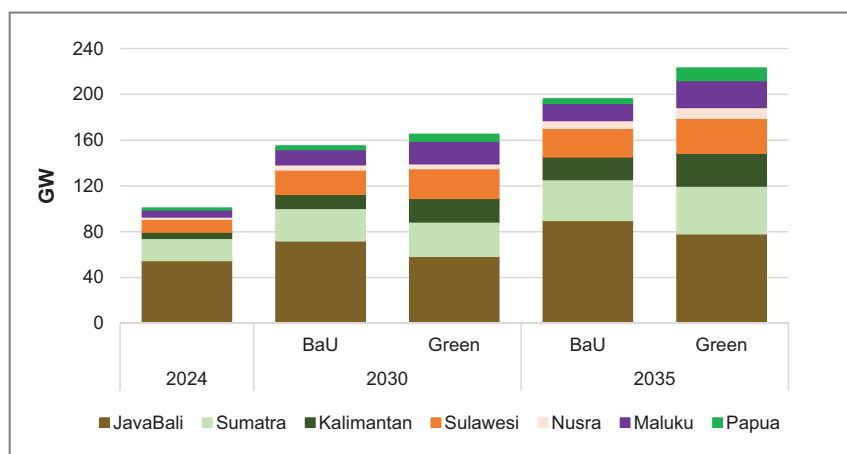


Figure 5.3 Regional Power Plant Capacity

5.4 REGIONAL RE POWER PLANT CAPACITY

In 2024, the majority of RE power plant capacity is in Sumatra (43%) and Java (37%). This pattern is expected to continue for the next decade despite of acceleration on additional capacity in several other regions. In particular, Maluku is projected to experience a very significant increase in RE capacity of around 32% per year, even though its share is only 1% of total RE power plant considering that 85% of power plant in the Maluku region are coal-fired power plants and diesel power plants. Therefore, the addition of RE power plants, especially solar power plants, will significantly impact the high growth of RE power plant. RE power plants in Maluku will increase from 9.2 MW in 2024 to 194 MW in 2035. A detailed RE capacity per region is presented in Figure 5.4.

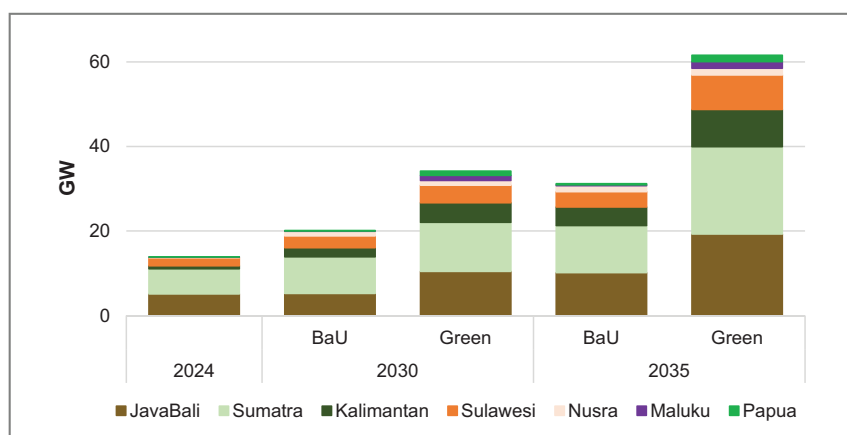


Figure 5.4 Regional RE Power Plant Capacity

5.5 PRIMARY ENERGY SUPPLY PER REGIONAL

In line with the development of final energy demand, Java-Bali is expected to continue to dominate primary energy supply until 2035 under both scenarios. In 2035, the primary supply share is projected to be around 42% in Java-Bali, 26% in Sumatera, 14% in Sulawesi, and 11% in Kalimantan. Meanwhile, the share in Nusa Tenggara, Maluku, and Papua collectively remains below 5%. A complete overview of primary energy supply by region is shown in Figure 5.5.

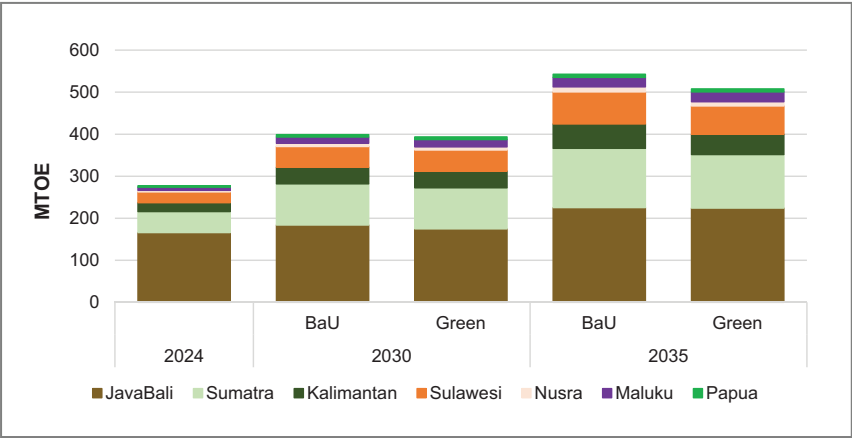


Figure 5.5 Primary Energy Supply per Region

CHAPTER 06

ANALYSIS OF ENERGY TRANSITION ACHIEVEMENT

CHAPTER 6

ANALYSIS OF ENERGY TRANSITION ACHIEVEMENT

Indonesia is committed to reduce GHG emissions to achieve the 2030 ENDC and 2060 NZE targets. Therefore, it takes sustainable energy transition efforts from the utilization of fossil energy to renewable energy and clean technology, considering that the energy sector will be the largest CO₂ emissions contributor. This chapter discusses an analysis of the achievements of energy transition programs and the required efforts.

6.1 BIOFUELS

The utilization of biofuels is one of the government's efforts to reduce GHG emissions in the energy sector, while reducing the dependency on fuel imports. In general, there are three types of biofuels. They are Biodiesel, Bioethanol, and Bioavtur. In order to support this development, the Ministry of Energy and Mineral Resources through the Directorate General of New Renewable Energy and Energy Conservation has stipulated Regulation of the Minister of Energy and Mineral Resources Number 4 of 2025 concerning Biofuels Business and Utilization. Furthermore, currently Directorate General of New Renewable Energy and Energy Conservation is drafting a ministerial decision concerning the phases of biofuel utilization which includes the determination of the development and utilization roadmap of the three types of biofuel by 2035.

6.1.1 Biodiesel

Since 2015, through the biofuels utilization mandatory phases, the implementation of biodiesel blending in diesel fuel in Indonesia has been continuously increased to 20% (B20) by 2016, 30% (B30) by 2020 and 35% (B35) by February 2023. Since January 2025, it has reached 40% (B40), the highest achievement in the world. The implementation of B35 Mandatory Program in 2024 successfully distributed biodiesel across Indonesia with a total volume of 13.15 million kL or approximately 98% of the total allocation

of 13.4 million kL. This program has a positive impact on various aspects, including foreign exchange savings of 7.86 billion USD or 124.28 trillion Rupiah, an increase in the added value of CPO into biodiesel of 17.68 trillion Rupiah, employment of more than 12 thousand people (off-farm) and 1.6 million people (on-farm), and a reduction in emissions of 34.93 million tons of CO₂-eq.

The government has set the implementation of diesel fuel with a 40% palm oil-based biodiesel blend (B40) starting on 1 January 2025. Prior to implementation, various technical tests were conducted, including a B40 road test for the automotive sector in 2022 and a B40 test for non-automotive diesel engines in 2024, covering heavy mining equipment, agricultural machinery, power plants, trains, and sea transportation. The test results showed that there were no significant obstacles in the use of B40 for these equipment.

The implementation of B40 Mandatory Program in 2025 sets a total biodiesel allocation of 15.6 million kL, consisting of 7.55 million kL for PSO and 8.07 million kL for non-PSO. This program is supported by 24 Biofuels Business Entities that will distribute biodiesel, 2 Fuel Business Entities that will distribute B40 for PSO and non-PSO, and 28 Fuel Business Entities that will distribute B40 specifically for non-PSO. With the B40 initiative, the government aims to expand the utilization of palm oil-based biodiesel across all sectors.

The government is currently conducting a comprehensive study on the implementation of B50, which will begin when all related instruments are ready. This is very important to consider since there are still technical complaints on the utilization of B40, such as a 26%-39% reduction in filter life in the automotive sector and an increase in specific fuel consumption (SFC) in the power generation sector. This leads to more wasteful fuel consumption. In addition, there are also other challenges such as a significant difference in viscosity compared to conventional diesel. Biodiesel has a viscosity of around 6 mm²/s, higher than in diesel of 4.5 mm²/s. This higher viscosity results in less optimal fuel atomization, resulting in slower combustion and increased fuel pump workload. A series of initial B50 tests will also be carried out in the near future to address these challenges, both in the automotive and non-automotive sectors. With the above considerations, the implementation of B-50, which is planned to be conducted in the second semester of 2026, needs more attention to be managed sustainably after the B-50 is established.

6.1.2 Bioethanol

Bioethanol mandatory program has not been implemented due to limited raw materials and the absence of an incentive mechanism provided by the government despite that since 2013 the Government has issued Presidential Regulation Number 40 of 2013 concerning the Acceleration of Bioethanol Provision for Biofuels. However, since July 2023, PT Pertamina Patra Niaga has conducted a market trial of Pertamina Green 95 (RON95 + RON98 + 5% Bioethanol). As a result, in 2024 its sales reached 7.5 thousand KL spread across 107 outlets. The results of this trial show several challenges in bioethanol development, including the high and fluctuating price of raw materials (especially molasses), and the need for an incentive mechanism to cover the price gap between Bioethanol market index price and conventional gasoline. On the supply side, the capacity of the FGE (Fuel Grade Ethanol) Bioethanol Plant in 2024 has reached 60 thousand KL/year, consisting of three plants with Biofuels business license and two plants without Biofuels business license. However, currently there are only two plants ready to produce FGE with a capacity of 36 thousand kL/year, namely PT Energi Agro Nusantara and PT Molindo Raya Industrial. Meanwhile, one business entity is committed to producing 20 thousand kL/year, namely PT Indonesia Ethanol Industri. The Ministry of Energy and Mineral Resources targets 5% mandatory bioethanol or E5 in the fuel blend by 2026 in the Java region by considering the ease of ethanol distribution.

From a technical perspective, one of the main obstacles in the utilization of bioethanol as a fuel is its low energy density compared to gasoline. With an energy value of only around 21–23 MJ/L, bioethanol produces less energy per liter than gasoline, which has an energy density of 32–34 MJ/L. Therefore, bioethanol-fueled vehicles tend to consume more fuel for the same distance, resulting in reduced operational efficiency. Despite of high octane number (RON) in bioethanol which allows it to be used in high-compression engines, such engine modifications require technological investment and design adaptations. In addition to its low energy density, bioethanol's ability to react with aluminum and damage its protective coating makes the engine more vulnerable to corrosion. However, the domestic industry is now technically capable of producing ethanol-fueled engines that can adapt to even 100% ethanol content.

Therefore, technical challenges must be a primary consideration in implementing the transition to bioethanol-based fuels to ensure efficiency and economics. On the other hand, the national implementation of bioethanol will significantly impact the energy mix by reducing gasoline's dominance as the most consumed fuel in the transportation sector.

6.1.3 Bioavtur

The world's aviation fuel demand increases by around 5% per year until 2030. Thus, the use of bioavtur will have an impact on increasing CO₂ emissions in the world aviation of around 682 million tons by 2030. On the global side, the development of Sustainable Aviation Fuel (SAF) is a commitment from the International Civil Aviation Organization (ICAO) to achieve zero carbon dioxide emissions in aviation by 2050. Mixing avtur with biofuel is expected to significantly reduce the total amount of emissions as the use of fuel is increasing.

Bioavtur development is currently only at the study and trial phase. In October 2023, a successful flight test of J2.4 fuel was conducted on PT Garuda commercial aircraft. In August 2025, PT Pertamina and its subsidiary, Pelita Air, conducted its inaugural flight from Jakarta to Bali using SAF derived from Used Cooking Oil (UCO). Indonesia is currently capable of producing bioavtur but not yet on a commercial scale. In 2021, PT Kilang Pertamina Internasional (PT KPI) produced 350,000 kl/year of bioavtur (J2.4). The raw material used is RBDPKO (Refined Bleached Deodorized Palm Kernel Oil), which is palm kernel oil that has been refined from CPKO (Crude Palm Kernel Oil). Some challenges in developing bioavtur include the limited availability of raw materials for SAF production due to potential overlap with food sector needs, as well as the need to reduce production costs to be comparable to fossil jet fuel.

Another challenge is the potential use of CPO as a bioavtur feedstock for international flights. It receives opposition from European countries. The current direction for bioavtur development is based on UCO (Used Coal). Prior to bioavtur implementation, an integrated waste cooking oil collection mechanism from households and commercial sector (hotels and restaurants) is needed to ensure a sustainable supply of bioavtur feedstock

6.2 HYDROGEN

Indonesia has significant potential for low-carbon hydrogen production, including grey, blue, and green hydrogen. Currently, hydrogen in Indonesia is only used in the industrial sector to produce ammonia, methanol, peroxide, and oil refineries. To accelerate the utilization of hydrogen as energy, the Directorate General of New Renewable Energy and Energy Conservation has developed standards for hydrogen fuel quality and product specifications, hydrogen refueling stations, and hydrogen system

safety. The Directorate General of New Renewable Energy and Energy Conservation has also published the National Hydrogen and Ammonia Roadmap (RHAN) in 2025 as a comprehensive guide for stakeholders in building and developing a sustainable hydrogen and ammonia ecosystem in Indonesia. RHAN includes measurable targets and action plans for the gradual development of hydrogen and ammonia in five-year intervals until 2060.

In general, hydrogen as an energy raw material can be used to substitute fossil fuels in the transportation, industrial, and power generation sectors. In the transportation sector, hydrogen is used as fuel for Fuel Cell-Based Vehicles (FCEVs). This includes cars, buses, and trucks. Hydrogen is also used in pilot projects for hydrogen-fueled ships and trains. Meanwhile, in the power generation sector, hydrogen is used in fuel cells to generate electricity and heat, especially in locations that require decentralized or inaccessible resources, such as in remote areas. Furthermore, hydrogen can be integrated into the electricity system by utilizing hydrogen turbines in gas-fired power plants as a co-firing of fossil fuels, or converted into ammonia and then burned in steam-fired power plants to produce steam that drives turbines.

The utilization of hydrogen as energy in Indonesia's transportation and industrial sectors could be one way to accelerate economic growth to 8% by 2029, increase the renewable energy mix, and reduce emissions in the energy sector. This is supported by several pilot projects that have been developed in Indonesia, including:

1. PLN's Hydrogen Refuelling Station (HRS) facilities in Senayan, Pertamina's HRS in Daan Mogot, and Toyota Motor Manufacturing Indonesia's HRS in Karawang.
2. Toyota Mirai and Hyundai Nexo are hydrogen-powered vehicles that have been produced and operated in several countries.
3. PT Inka has completed a pilot project for a hydrogen train with a power capacity equivalent to 10 kW.
4. PLN Indonesia Power's 7% hydrogen co-firing trial at the Pesanggaran Diesel Power Plant in Bali.

However, hydrogen development faces regulatory, economic, technological, and infrastructure challenges. Policy uncertainty at the national and regional levels can hamper the development of hydrogen projects. Furthermore, the cost of producing green hydrogen through electrolysis remains high along with the high price of electricity from renewable energy plants, making it difficult to compete with cheaper gray hydrogen

produced from natural gas without CCS. However, larger-scale production and declining technology prices, including electrolyzers, which are expected to drop by up to 30% in some key markets, are expected to reduce hydrogen production costs. For example, in 2020, the global production cost of green hydrogen was 2-8 times higher than gray hydrogen, ranging from 2-8 USD/kg (IEA, 2021). Meanwhile, the Hydrogen Council (2021) projected that by 2030, the production cost of green hydrogen is expected to drop by 62%, reaching breakeven with gray hydrogen. Establishing a minimum carbon price of 50 USD could accelerate cost parity by up to two years, depending on regional gas and renewable energy prices (Hydrogen Council, 2021). On the other hand, based on the ERIA study in 2025, the main component of green hydrogen production costs comes from renewable energy-based electricity costs with a tariff of around 0.03–0.05 USD/kWh and will result in a green hydrogen production cost of 2.20–3.50 USD/kg-H₂. Thus, the economics of hydrogen are highly dependent on electricity prices and process efficiency, as well as supporting distribution infrastructure. To lower the price of landed hydrogen and accelerate commercialization, policy interventions such as access to competitive electricity tariffs and fiscal incentives are needed. This estimate includes the entire production process, but does not include additional costs for upgrading port facilities such as loading and unloading equipment and others. For comparison, based on PLN data in 2024, the global selling price of hydrogen ranges from 5–7 USD/kg, with the lowest price in China of around 5–15 USD/kg, and the highest in Japan of around 10–15 USD/kg. Therefore, based on the study, PLN proposed that the selling price of hydrogen in Indonesia should reach 8 USD/kg to enable a sustainable low-carbon hydrogen business chain in Indonesia.

6.3 NUCLEAR POWER PLANT

The Government Regulation on National Energy Policy has positioned nuclear energy as a counterbalance and strategic option in decarbonization efforts. The National Energy Policy targets Nuclear Power Plants to be operational in Indonesia by 2032 with a capacity of 250 megawatts, and will continue to grow to reach 45 gigawatts by 2060. In addition, nuclear has been included in planning documents such as the RPJPN (National Long Term Development Plan) 2025-2045, RUKN (National Electricity General Plan) 2025-2060, and PLN RUPTL (Power Supply Business Plan) 2025-2034. The RPJPN document explains that the development of Indonesia's nuclear energy is supported by the potential of uranium and thorium as abundant nuclear compositions which are concentrated in Kalimantan and Papua. Currently, nuclear technology has entered III/III+ and IV generation and a number of 33 countries have utilized nuclear power plants.

Of the five most populous countries, it is only Indonesia which has not developed large-scale nuclear power plant. RUKN targets the construction of a nuclear power plant of around 7 GW by 2040. PLN has stated its readiness with the initial plan for the construction of a nuclear power plant scheduled to begin in 2032 with a capacity of around 2 x 500 MW projected to be located in Bangka Belitung and/or West Kalimantan.

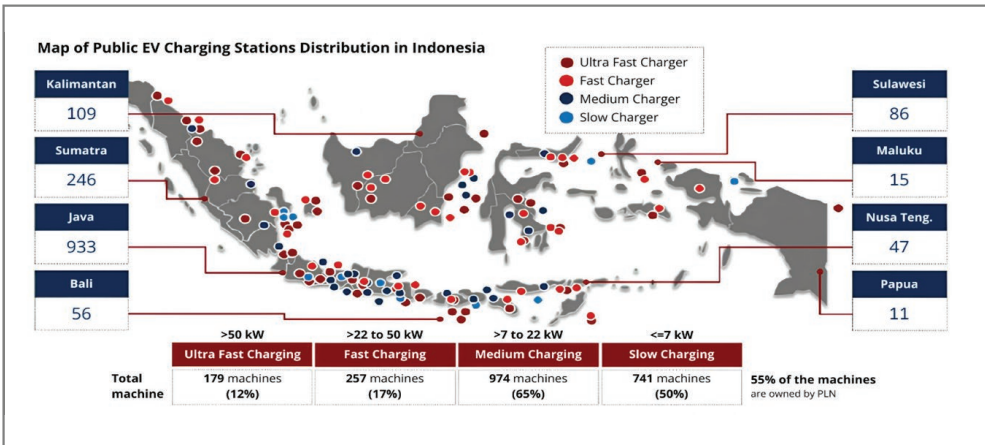
To achieve this plan, the policy and institutional foundations have been systematically established since 2021. It begins with the establishment of the NEPIO (Nuclear Energy Program Implementing Organization) Preparation Team, which later evolved into a proposed Nuclear Energy Program Implementation Committee (KPPEN) to coordinate national preparations. In November 2025, the NEPIO Presidential Regulation was completed, harmonized, and ready for immediate promulgation. Furthermore, the Government, through the National Agency for Research and Innovation (BRIN), has also conducted various studies to identify several potential locations for nuclear power plants throughout Indonesia. However, it must be acknowledged that nuclear power plant development is not without challenges, particularly in terms of funding and construction time. The investment cost for a single nuclear power plant unit can reach 3.8 billion USD, with a construction period of approximately 4-5 years and a preparation period of at least 3 years. Therefore, if a nuclear power plant is to be built, it will require 8-10 years from preparation to COD. Furthermore, public concerns about the risk of natural disasters are also a serious concern for the government. Nuclear power plants require relatively high safety standards, including in reactor cooling systems and waste management. The government needs to pay close attention to mitigation and strict supervision, as well as international cooperation to ensure operations through BAPETEN. Furthermore, local human resource readiness is still limited. Therefore, education and training programs in the nuclear energy sector must be strengthened. These various aspects are crucial considering that the National Energy Policy targets the first nuclear power plant (NPP) to be operational by 2032, which is only about seven years from now. Practically, a viable alternative is to build a Small Modular Reactor (SMR), which has a simpler construction process than a conventional nuclear power plant. However, further preparation is needed considering that the 2 GW nuclear power plant is targeted to be operational by 2035. Therefore, a serious commitment from the government and accelerated efforts are needed so that nuclear power plant construction can begin immediately and operate according to the roadmap in National Energy Policy.

6.4 ELECTRIC VEHICLE

As one of energy transition programs, the development of electric vehicles in Indonesia has shown significant progress by 2025. The government, through the Ministry of Energy and Mineral Resources, has set an ambitious target for the population of battery-based electric vehicles (KBLBB) operating on roads by 2030, totaling 2 million electric cars and 13 million electric motorcycles. This step is not only intended to reduce dependence on fossil fuels but also serves as part of a strategy to support Indonesia's commitment to reducing GHG emissions.

To ensure the achievement of the target, the government is also preparing the supporting infrastructure in the form of Public Electric Vehicle Charging Stations (SPKLU). The Minister of Energy and Mineral Resources Decision Number 24.K/TL.01/MEM.L/2025 includes SPKLU development plan from 2025 to 2030. By the end of that period, Indonesia targets the construction of 62,918 SPKLU with a charging technology composition consisting of 55% of medium chargers, 28% of fast chargers, and 17% of ultra-fast chargers. The government also regulates the distribution of SPKLU construction with a certain ratio, for example, five to one in the Greater Jakarta area and twelve to one outside Greater Jakarta. SPKLU construction locations are planned to be spread across shopping centers, offices, toll road rest areas, gas stations, hospitals, railway stations, bus terminals, hotels, and ports. Thus, electric vehicle users are expected to experience comfort and ease in accessing charging facilities at various strategic locations.

Based on its achievements, the development of the electric vehicle ecosystem also shows a positive trend. As of July 2025, there were 4,186 operating charging stations (SPKLU) at 2,789 locations, along with 1,902 battery exchange facilities (SPBKLU) (Figure 6.1). The electric vehicle population has reached 252,000 units, consisting of 184,000 electric motorcycles, 67,000 passenger cars, and the remainder buses and commercial vehicles. Compared to 2024 with the total of 207,000 units, the number in 2025 represents a 78% increase. Comparing 2023 and 2022, the growth has tripled. This growth reflects public acceptance of electric vehicles as an environmentally friendly transportation alternative. The significant increase in EVs over the past two to three years requires support from the domestic industry, particularly the battery industry, whose raw materials are abundantly available in Indonesia.



Source : Coordinating Ministry for Infrastructure and Regional Development; PLN, 2024

Figure 6.1 Distribution of SPKLU in Indonesia

Several barriers in EVs utilization in the transportation sector include limited charging infrastructure, vehicle travel distance, and operational costs. Although incentives have lowered the price of electric vehicles, potential buyers still consider performance, vehicle options, and after-sales services. However, to support the massive development of electric vehicles, strategic supply chain development is needed. It can be reached by developing the electric vehicle industry specifically in investment of strategic supply chain such as battery cell manufacturers as well as strengthening the electric vehicle support ecosystem.

CHAPTER 07

CLOSING

CHAPTER 7

CLOSING

7.1 CONCLUSION

Indonesia's efforts towards the energy transition era by 2035 as depicted in the Green scenario need to be supported by:

1. Energy efficiency in various sectors

Indonesia needs to strengthen energy efficiency across all sectors to support the energy transition. These efforts include replacing equipment with energy-saving appliances, electrifying industrial processes, and adopting electric vehicles in the transportation sector to reduce the dominance of fossil-fueled vehicles.

2. Optimization of bioethanol use

The use of bioethanol contributes to increasing the renewable energy mix, but is currently hampered by custom issues and blending facility permit requirements. Inter-ministerial policy coordination is needed to overcome these obstacles and support the achievement of the 19% to 23% renewable energy target by 2030.

3. The development of solar power plant ecosystem

The dominance of the development of the solar power plant ecosystem is important to achieve the projected generation capacity of 16.5 GW by 2035. The sustainability of solar power plant growth will be stronger if it is supported by the development of the domestic solar cell industry to reduce dependence on imports, to lower the costs, and to create more jobs.

4. Electricity consumption per capita

Electricity consumption per capita as an important indicator of a country's level of welfare, economic progress and energy development, requires careful planning of power plants and transmission networks development.

5. The utilization of low carbon technology in mineral downstreaming

Mineral downstreaming and smelter industry expansion are key drivers of regional economic growth (particularly in Sulawesi and Maluku). Considering that the

majority of smelter industry energy is currently coal-based, a gradual substitution to low-carbon energy sources and the adoption of low-emission technologies is necessary to support industrial decarbonization.

6. Energy consumption and production pattern per region

The projections of energy consumption and production in Indonesia for the 2024-2035 period show a consistent pattern in which the energy consumption and production domination lies in the Java-Bali region, in line with the high concentration of population and economic activity. However, along with industrial development and government policies, regions outside Java-Bali, such as Maluku, Sulawesi, and Kalimantan, are showing significant growth in energy consumption, particularly in mineral processing and smelting industries.

7.2 RECOMMENDATION

1. Strengthening cross sectors coordination

- a. A clear coordination mechanism between ministries and stakeholders is needed to resolve crucial policy obstacles such as exemption/adjustment of ethanol custom duty and simplification of blending permits as well as policies that support local content in the solar power plant industry.
- b. A need to synchronize national targets with the industry roadmap and fiscal incentives to accelerate the realization of RE targets.

2. Accelerate infrastructure development and enhance national industry capacity

- a. The government needs to prioritize investment in the solar power plant value chain (solar cells, modules, inverters, and supporting components) to increase domestic production capacity and reduce dependence on imports.
- b. The government needs to facilitate access to financing for renewable energy projects and industrial electrification projects.

3. Promote demand shifting and electrification

- a. The government needs to implement programs to increase energy efficiency, industrial process electrification, and electric vehicles (EVs) acceleration by determining efficiency standards, large-scale pilot programs, and incentives for EVs.
- b. The government needs to develop incentives for the use of biofuels and the integration of renewable energy sources into the electricity system to reduce emissions in the transportation and industrial sectors.

4. Decarbonization in “energy consuming” industry
 - a. It is necessary to prepare a program for implementing low-carbon technology in mineral downstreaming and smelters, including substituting coal with low-emission alternatives and increasing process efficiency.
 - b. It is necessary to design incentive and regulatory schemes to encourage clean technology investment in the developing industrial zones.
5. A consistence and transparent investment climate
 - a. The government needs to emphasize its commitment to issue stable regulations and long-term incentives to attract businesses, both state-owned enterprises and private sector, to invest in low-carbon technology and renewable energy.
 - b. The government needs to ensure that fiscal and non-fiscal policies support the achievement of the 2060 Net Zero Emissions (NZE) target through roadmaps, monitoring mechanisms, and transparency of emissions reporting targets.

ATTACHMENT

ATTACHMENT 1

OUTLOOK SUMMARY

No	Energy Indicator	Unit	2024	National Projection of BaU Scenarios		National Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	189	272	363	268	319
2	Final Energy Demand for Electricity	TWh	417	580	757	609	822
3	Primary Energy Supply	MTOE	277	399	543	393	508
4	Power Plant Capacity	GW	101	155	196	165	223
5	Electricity Production	TWh	372	644	824	676	898
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	738	1,029	1,395	950	1,110

No	Energy Indicator	Unit	2024	Sumatra Projection of BaU Scenarios		Sumatra Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	40	64	91	63	73
2	Final Energy Demand for Electricity	TWh	87	134	185	130	168
3	Primary Energy Supply	MTOE	50	98	141	98	127
4	Power Plant Capacity	GW	19	28	36	30	42
5	Electricity Production	TWh	71	149	195	143	194
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	120	218	314	190	212

No	Energy Indicator	Unit	2024	Java-Bali Projection of BaU Scenarios		Java-Bali Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	101	127	151	130	157
2	Final Energy Demand for Electricity	TWh	187	245	303	283	409
3	Primary Energy Supply	MTOE	166	184	225	175	224
4	Power Plant Capacity	GW	55	72	90	59	78
5	Electricity Production	TWh	200	287	341	244	330
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	453	505	614	425	513

No	Energy Indicator	Unit	2024	Kalimantan Projection of BaU Scenarios		Kalimantan Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	19	30	43	27	30
2	Final Energy Demand for Electricity	TWh	24	33	43	33	42
3	Primary Energy Supply	MTOE	21	40	58	40	48
4	Power Plant Capacity	GW	5	12	20	21	29
5	Electricity Production	TWh	18	51	75	79	101
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	48	91	135	98	100

No	Energy Indicator	Unit	2024	Sulawesi Projection of BaU Scenarios		Sulawesi Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	19	35	56	32	40
2	Final Energy Demand for Electricity	TWh	68	99	135	91	114
3	Primary Energy Supply	MTOE	26	49	76	50	67
4	Power Plant Capacity	GW	11	21	25	26	31
5	Electricity Production	TWh	46	81	104	105	138
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	73	136	213	141	169

No	Energy Indicator	Unit	2024	Nusra Projection of BaU Scenarios		Nusra Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	3	4	6	5	6
2	Final Energy Demand for Electricity	TWh	8	12	17	13	18
3	Primary Energy Supply	MTOE	4	8	12	7	10
4	Power Plant Capacity	GW	2	5	7	4	9
5	Electricity Production	TWh	6	18	29	14	18
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	9	16	25	14	18

No	Energy Indicator	Unit	2024	Maluku Projection of BaU Scenarios		Maluku Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	4	8	12	7	9
2	Final Energy Demand for Electricity	TWh	28	39	51	38	47
3	Primary Energy Supply	MTOE	7	15	22	18	23
4	Power Plant Capacity	GW	6	13	15	20	23
5	Electricity Production	TWh	26	47	66	77	99
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	28	51	78	70	86

No	Energy Indicator	Unit	2024	Papua Projection of BaU Scenarios		Papua Projection of Green Scenarios	
				2030	2035	2030	2035
1	Final Energy Demand	MTOE	3	4	5	4	4
2	Final Energy Demand for Electricity	TWh	14	19	23	20	24
3	Primary Energy Supply	MTOE	3	6	8	6	7
4	Power Plant Capacity	GW	2	3	4	6	11
5	Electricity Production	TWh	4	11	14	14	17
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	7	12	15	12	12

ATTACHMENT 2

CONVERSION FACTOR

Energy	Unit	Multiplying Factor to BOE (Barrel Oil Equivalent)
Coal		
Anthracite	Ton	4.9893
Imported Coal	Ton	4.2766
Kalimantan Coal	Ton	4.2
Ombilin Coal	Ton	4.8452
Tanjung Enim Coal	Ton	3.7778
Lignite	Ton	3.0649
Riau Peat	Ton	2.5452
Briquettes	Ton	3.5638
Biomass		
Charcoal	Ton	4.9713
Firewood	Ton	2.2979
Natural Gas	MSCF	0.1796
Gas products		
City Gas	Thousand Kcal	0.0007
CNG	Thousand Kcal	0.0007
LNG	Ton	8.0532
LNG	MMBTU	0.1796
LPG	Ton	8.5246

Energy	Unit	Multiplying Factor to BOE (Barrel Oil Equivalent)
Oil		
Condensate	Barrel	0.9545
Petroleum	Barrel	1
Fuel		
Aviation Gasoil (Avgas)	Kilo Liter	5.553
Aviation Turbine Gas (Avtur)	Kilo Liter	5.8907
Super TT	Kilo Liter	5.8275
Premix	Kilo Liter	5.8275
Premium	Kilo Liter	5.8275
Kerosene	Kilo Liter	5.9274
Diesel	Kilo Liter	6.4871
MDF	Kilo Liter	6.6078
FO	Kilo Liter	6.9612
Oil Products		
Other oil products	Barrel	1.02
Refinery Fuel		
Refinery Fuel Gas (RFG)	Barrel	1.6728
Refinery Fuel Oil (RFO)	Barrel	1.1236
Feed Stock	Barrel	1.0423
Power Plant	MWh	0.613

ATTACHMENT 3

EMISSION FACTOR

No.	Type of Energy	Net Energy Content	% Carbon Content	CO ₂ (Kg CO ₂ /TJ)
1	Gas	0	73.4	57,600
2	Gasoline	5.8	84.6	73,290
3	Avtur	5.9	85	71,348
4	Kerosene	5.9	85	72,430
5	Diesel	6.5	86.5	76,000
6	LPG	5.6	82	59,500
7	Coal	4	74.6	100,000
8	Briquettes	3.5	74.6	100,000
9	Avgas	5.6	84.6	71,348
10	Fuel Oil	7	86.5	77,900
11	DME	5.6	82	57,600

ATTACHMENT 4

GLOSSARY

Avgas

A type of aviation fuel used in piston engine aircraft.

Avtur (Aviation Turbine Fuel)

A type of aviation fuel designed for gas turbine engines-aircraft.

Coal

Sedimentary rocks that can burn, are formed from organic deposits, mainly plant remains and are formed through the process of coalescence.

Bioavtur

Aviation fuel produced by blending or processing vegetable oils, animal-fats, or non-fossil biomass in gas turbine engines to reduce emission.

Biomass

Renewable energy from organic materials such as animals and plants.

Biofuel

Any fuel, whether solid, liquid or gas, that is produced from organic materials which may be derived directly from plants or indirectly from industrial, commercial, domestic or agricultural waste.

B40

The result of mixing diesel oil with 40% vegetable oil derived from palm oil or Crude Palm Oil (CPO).

Captive Power

A power plant owned and operated by a company or industry to meet its own electricity needs, not for sale to the public grid.

Co-firing

Mixing biomass with coal to be burned together in a single steam-Coal Fired Power Plant (PLTU).

RE (Renewable Energy)

Energy that comes from natural processes that is produced continuously and sustainably without having to wait for millions of years like fossil energy.

Carbon Emission

The remaining results of fuel combustion in internal combustion engines, external combustion engines, or jet engines that are released through the engine exhaust system.

Final Energy

Energy that can be directly consumed by end users.

Primary Energy

The energy that is first consumed in an area, both which comes directly or indirectly from nature.

Gas

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the form of a gas phase obtained from the oil and gas mining process.

Industry

A field or economic activity related to the processing/manufacture of raw materials or finished goods in factories using skills and labor and the use of tools in the field of processing agricultural products and their distribution as the main activity.

KEN (*Kebijakan Energi Nasional*/National Energy Policy)

Energy management policies based on just, sustainable and environmentally friendly principles to create national energy independence and security.

Commercial

An activity carried out with the aim of gaining profit, generally in the form of activities related to commerce.

LEAP (Low Emission Analysis Platform)

An energy planning simulation model that is capable of analyzing energy from demand to supply in an integrated manner.

Oil

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the liquid or solid phase.

Power Plant

A collection of equipment and machines used to generate electricity through the process of transforming energy from various energy sources.

Region

Regional division based on geographic proximity used in energy modeling (LEAP).

Household

Consists of one or several people who live in the same residence which include one family or group of people.

Other Sectors

Several activities that correlate with supporting commercial businesses consist of agricultural, construction and mining activities.

Scenario

Conditions or circumstances that may occur based on the assumptions input into energy modeling.

Solar Cell

A semiconductor device that converts sunlight energy into electrical energy through a process called the photovoltaic effect.

Solar Farm

A large-scale solar power plant consisting of an array of solar panels installed in open areas to capture solar radiation and convert it into electrical energy.

Transportation

One part of the goods and passenger distribution network that has developed very dynamically and plays a role in supporting political, economic, socio-cultural as well as defense and security development.

NZE (Net Zero Emission)

Conditions where the amount of carbon emissions released into the atmosphere does not exceed the amount of emissions that the earth can absorb.

Gross Domestic Product

The value of all goods and services produced by a country in a certain period.

Projection

Projected conditions that may occur at a certain time based on the assumptions used.

Roadmap

A strategic plan that describes the stages, targets, and implementation steps to achieve specific timeframe, for example in energy or technology development.

ATTACHMENT 5

ABBREVIATIONS

APBN	: <i>Anggaran Pendapatan dan Belanja Negara</i> (State Budget)
APK DEN	: <i>Anggota Pemangku Kepentingan Dewan Energi Nasional</i> (Members of National Energy Council from Stakeholders)
Bappenas	: <i>Badan Perencanaan Pembangunan Nasional</i> (National Development Plan Agency)
BBM	: <i>Bahan Bakar Minyak</i> (Fuel)
BBN	: <i>Bahan Bakar Nabati</i> (Biofuel)
BOE	: Barrel Oil Equivalent
BPP	: <i>Biaya Pokok Penyediaan</i> (Basic Cost of Supply)
BPS	: <i>Badan Pusat Statistik</i> (Statistics Indonesia)
BTS	: Base Transceiver Station
CCS	: Carbon Capture and Storage
CO	: Carbon Dioxide
CO ₂ eq	: Carbon Dioxide equivalent
DEN	: <i>Dewan Energi Nasional</i> (National Energy Council)
Dj EBTKE	: <i>Direktorat Jenderal Energi Baru Terbarukan dan Konservasi Energi</i>
DME	: Dimethyl Ether
EV	: Electric Vehicle
EBT	: <i>Energi Baru Terbarukan</i> (New Renewable Energy)
EBTKE	: <i>Energi Baru Terbarukan dan Konservasi Energi</i> (New Renewable Energy and Energy Conservation)
GRK	: <i>Gas Rumah Kaca</i> (Green House Gas)
GSEN	: <i>Grand Strategi Energi Nasional</i> (National Energy Grand Strategy)
Gt	: Giga tonne
GW	: Giga Watt
HEESI	: Handbook of Energy & Economic Statistics of Indonesia
H ₂	: <i>Hidrogen</i> (Hydrogen)
IEA	: International Energy Agency
IKN	: <i>Ibu Kota Negara</i> (Capital City)
KBLBB	: <i>Kendaraan Bermotor Listrik Berbasis Baterai</i> (Battery based electric vehicles)
KEN	: <i>Kebijakan Energi Nasional</i> (National Energy Policy)

KESDM	: <i>Kementerian Energi dan Sumber Daya Mineral</i> (Ministry of Energy and Mineral Resources)
kW	: kilo Watt
kWh	: kilo Watt hour
LEAP	: Low Emissions Analysis Platform
LPEM	: <i>Lembaga Penyelidikan Ekonomi dan Masyarakat</i> (Institute for Research on Economics and Society)
LPG	: Liquefied Petroleum Gas
LNG	: Liquefied Natural Gas
LTSHE	: <i>Lampu Tenaga Surya Hemat Energi</i> (Energy Saving Solar Lights)
MMBTU	: Million British Thermal Unit
MT	: Metrik Ton
MTPA	: Metric Ton per Annum
MMLTPA	: Million Metric Ton per Annum
MVA	: Mega Volt Ampere
MW	: Mega Watt
MWh	: Mega Watt hour
NEPIO	: Nuclear Energy Program Implementing Organization
NDC	: Nationally Determined Contributions
Nusra	: Nusa Tenggara
NZE	: Net Zero Emission
OEI	: <i>Outlook Energi Indonesia</i> (Indonesia Energy Outlook)
PSO	: Public Service Obligation
PLN	: <i>Perusahaan Listrik Negara</i> (Electricity State Own Enterprise)
Perpres	: <i>Peraturan Presiden</i> (Presidential Regulation)
PLTA	: <i>Pembangkit Listrik Tenaga Air</i> (Hydro Power Plant)
PLTB	: <i>Pembangkit Listrik Tenaga Bayu</i> (Wind Power Plant)
PLTD	: <i>Pembangkit Listrik Tenaga Diesel</i> (Diesel Power Plant)
PLTG	: <i>Pembangkit Listrik Tenaga Gas</i> (Gas Power Plant)
PLTM	: <i>Pembangkit Listrik Tenaga Minihidro</i> (Minihydro Power Plant)
PLTMH	: <i>Pembangkit Listrik Tenaga Mikrohidro</i> (Microhydro Power Plant)
PLTN	: <i>Pembangkit Listrik Tenaga Nuklir</i> (Nuclear Power Plant)
PLTP	: <i>Pembangkit Listrik Tenaga Panas Bumi</i> (Geothermal Power Plant)
PLTS	: <i>Pembangkit Listrik Tenaga Surya</i> (Solar Power Plant)
PLTU	: <i>Pembangkit Listrik Tenaga Uap</i> (Coal Fired Power Plant)
PP	: <i>Peraturan Pemerintah</i> (Government Regulation)
Pusdatin	: <i>Pusat Data dan Teknologi Informasi</i> (Data and Information Technology Center)

RUPTL	: <i>Rencana Usaha Penyediaan Tenaga Listrik</i> (Power Supply Business Plan)
RUEN	: <i>Rencana Umum Energi Nasional</i> (National Energy General Plan)
RON	: Research Octane Number
Setjen/SJ DEN	: <i>Sekretariat Jenderal Dewan Energi Nasional</i> (Secretariat General of the National Energy Council)
SPBU	: <i>Stasiun Pengisian Bahan Bakar Umum</i> (Fuel Filling Stations)
SPBKLU	: <i>Stasiun Penukaran Baterai Kendaraan Listrik Umum</i> (Public Electric Vehicle Battery Swap Station)
SPKLU	: <i>Stasiun Pengisian Kendaraan Listrik Umum</i>
TOE	: Tonne of Oil Equivalent
TSCF	: Trillion Standard Cubic Feet
TWh	: Terra Watt hour
UI	: Universitas Indonesia
USD	: United States Dollar



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